

SignalShap: Exact-Enumeration Ranking-Stage Attribution of Sources in Hybrid Recommenders

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ABSTRACT

Source ablation and Shapley attribution answer different questions: removal effect and credit allocation. Conflating them can select the wrong decision instrument.

SignalShap formulates ranking-stage source attribution in hybrid recommenders as a five-player cooperative game comprising collaborative filtering, content, popularity, recency, and sequential signals. The payoff is normalised discounted cumulative gain at rank 10 (NDCG@10) on a fixed, coalition-independent candidate set. Exhaustive evaluation of all $2^5 = 32$ coalitions yields exact Shapley values without coalition-sampling error. Six analytic games with known solutions are recovered exactly, with a maximum efficiency residual of 6.9×10^{-18} .

Across MovieLens-1M, Amazon Video Games, and a repeat-masked Gowalla stress test, MovieLens-1M and Gowalla each contain a source for which Shapley and ranking-stage leave-one-out disagree in sign by a material margin. Over ten stochastic fits, ranking-stage leave-one-out agrees more closely than Shapley with observed single-source end-to-end removal losses on every corpus: mean Kendall rank agreement is 0.96–1.00 versus 0.20–0.80, and the lowest-removal-loss source is identified on 90–100% versus 0–50% of fits. *SignalShap* therefore supports credit allocation within the declared ranking-stage game, whereas leave-one-out estimates removal effect; *SignalShap* is not a retirement rule. Candidate recall limits applicability: only MovieLens-1M clears the prespecified 0.60 gate under the primary configuration, Amazon Video Games clears it only in a larger-cap sensitivity, and repeat masking restricts Gowalla to relative robustness evidence.

1. Introduction

A production hybrid recommender is an ensemble of architecturally distinct scorers. A collaborative-filtering component exploits co-occurrence, a content component exploits item metadata, a popularity component exploits global frequency, a recency component exploits temporal locality, and a sequential component exploits order. Two questions then arise: how should the system's observed value be divided among its sources, and what would happen if one source were retired? These are credit-allocation and removal-effect questions, respectively, and they require different instruments. In practice, removal effects are commonly estimated by *leave-one-out* (LOO) ablation: disable one source, measure the change in ranking quality, and assign that change to the source.

LOO is meaningful only relative to the game on which it is computed, and this paper distinguishes two forms (Section 2.2). End-to-end LOO removes a source from retrieval and ranking together and is therefore the deployment intervention. Ranking-stage LOO removes the source only from a fixed-candidate ranker and measures a different, cheaper estimand. Dependence among sources does not invalidate either removal estimand: the intervention cost remains well defined for the interacting system being measured.

LOO is not, however, a complete rule for dividing credit. Hybrid sources often overlap: for example, implicit-feedback collaborative filtering and popularity can encode some of the same signal. When two sources are mutually substitutable, removing either alone may change little, assigning both near-zero conditional value even though their shared capability contributes to the system. A credit-allocation rule must also consider coalitions in which no substitute is available. Neither estimand dominates; they answer different questions, and this paper measures where they diverge.

Cooperative game theory provides such an allocation rule. The Shapley value [1] is the unique credit assignment satisfying efficiency, symmetry, null-player, and additivity, and it evaluates each player against *every* coalition rather than only against the grand coalition. Its adoption in machine learning has been broad [2, 3], most commonly at the granularity of *input features*, though component-level, neuron-level and data-level games also appear (Section 2.1). Here the players are *architectural sources*, and the payoff is a ranking metric.

Figure 1 summarises the pipeline. Moving from hundreds of features to five sources provides two practical advantages. First, generic model-agnostic feature Shapley is usually approximated because its coalition lattice is intractable; methods such as KernelSHAP therefore sample and inherit estimator variance, although exact algorithms exist for particular model classes, notably trees [4]. Five sources produce only $2^5 = 32$ coalitions, and each source has $2^4 = 16$ marginal contributions. We evaluate the complete game by exhaustive enumeration, so the attribution

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introduces no coalition-sampling error. Second, the small player set improves *verifiability*: we can construct games and interventions whose attributions must satisfy known constraints, then test whether the implementation does so. Section 4.1 applies this strategy to analytic and real fitted games.

Accordingly, this study develops and verifies an exact source-level attribution framework and asks when credit allocation and removal effect diverge. The objective is not to replace ablation with Shapley attribution, but to identify which estimand is appropriate for a given system-level decision.

1.1. Contributions

- **C1.** A precisely declared ranking-stage cooperative game over heterogeneous recommender sources, with a coalition-independent candidate set, a per-user formulation, and exact enumeration for small source sets (Section 3.1).
- **C2.** A formal characterisation of source allocation under redundancy, including efficiency and per-user decomposition, a counterexample to unconditional redundancy-positivity, and a corrected two-sided ϵ -substitutability result (Section 3.3).
- **C3.** An implementation-verification protocol combining analytic games with known Shapley vectors, controlled duplicate injection, and efficiency, null-player, and additivity diagnostics (Sections 4.1 and 4.2).
- **C4 (primary).** An empirical comparison of three games (refitted-head ranking-stage, fixed-head ranking-stage, and end-to-end retrieval) and two attribution rules across MovieLens-1M, Amazon Video Games, and a repeat-masked Gowalla stress test. The comparison separates credit allocation from removal effect, evaluates both against observed end-to-end source removals, and uses candidate recall to delimit where absolute attribution magnitudes are interpretable (Sections 4.3 and 4.7).

Scope SignalShap explains credit within a declared cooperative game; it is neither a causal attribution method nor a source-retirement rule. The paper makes no state-of-the-art accuracy claim against LightGCN or SASRec, which serve only as contextual references, and it does not claim that attribution-derived fusion improves ranking. The latter is reported as a negative result in Appendix A rather than as a contribution.

1.2. Research questions

RQ1 Does a fixed-candidate, coalition-independent cooperative game over architectural sources admit an exact additive decomposition of its value relative to baseline, and does the implementation satisfy known Shapley constraints on analytic and real fitted games?

RQ2 Does source-level Shapley attribution differ materially from leave-one-out ablation, and is the difference associated with source-score overlap or negative pairwise interaction under the declared game?

RQ3 How do attributions change across refitted-head ranking-stage, fixed-head ranking-stage, and end-to-end games, and does ranking-stage LOO or Shapley better predict observed end-to-end retirement cost?

Roadmap Section 2.1 positions the work against component-level and ranking-specific attribution. Section 3 defines the cooperative game, algorithm, formal properties, datasets, and evaluation protocol. Section 4 verifies the implementation and reports the attribution, sensitivity, and source-removal experiments. Sections 5 and 6 discuss the scope and conclusions; Appendix A reports the negative fusion result.

2. Literature review

2.1. Related work

Shapley values in machine learning SHAP [2] unified additive feature attribution; extensions cover trees [4], data valuation [5], and federated contribution [6]. Surveys [3, 7] note that model-agnostic estimators commonly sample, because the coalition lattice is exponential in the number of players, with exact procedures available for structured model classes. Our setting sidesteps this: five players, 32 coalitions, exact enumeration.

Feature-level XAI versus source-level attribution SHAP, LIME [8], and Integrated Gradients [9] answer “which input features moved this prediction?”. We answer “which architectural component earned the system’s ranking quality?”. The unit of attribution differs, the payoff is a ranking metric rather than a scalar output, and the actionable decision differs: ours is the allocation of system value across subsystems.

Attribution to model components and to performance Closer to our unit of analysis than feature-level XAI is work attributing behaviour to *model components* rather than inputs. Neuron Shapley [10] treats individual neurons as players with a performance payoff, and ensemble games [11] treat whole classifiers as players and allocate ensemble accuracy among them. This is the closest structural precedent for what we do, since both take trained models as players and a performance metric as the characteristic function. Related work attributes changes in *performance* rather than individual predictions [12, 13]. Component-level valuation therefore provides a direct precedent; this paper extends that unit of analysis to architectural sources in a ranking system.

The closest recent comparison to our allocation-removal distinction comes from ensemble forecasting: Kim et al. [14] compare leave-one-model-out importance with an all-subsets Shapley measure and relate their disagreement to overlap in model errors. Bell et al. [15] develop efficient

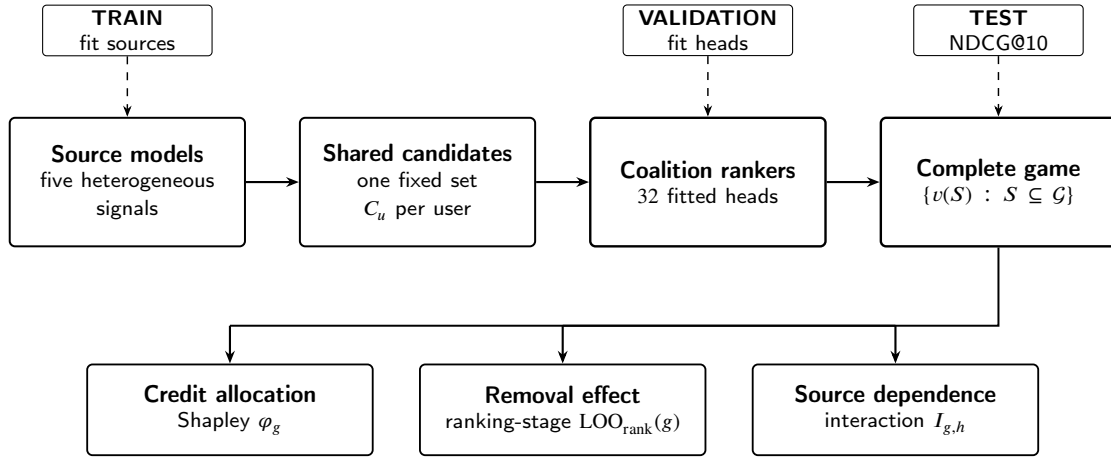


Figure 1: SignalShap workflow. Training data fit five source models; one coalition-independent candidate set is shared by every coalition. Validation data fit the coalition-specific heads, and test NDCG@10 defines the complete game. Exact enumeration supports three distinct diagnostics: Shapley credit allocation, ranking-stage leave-one-out removal effect, and pairwise source interaction.

Shapley performance attribution for least-squares models. Thus neither component-level performance attribution nor the contrast with a grand-coalition marginal is new by itself. Our narrower contribution is the fixed-candidate ranking game and its evaluation in a two-stage recommender.

Shapley beyond supervised prediction Shapley attribution has also been applied to unsupervised settings, where the absence of labels removes the natural payoff that classification or regression supplies. Our own earlier work [16] attributes cluster structure to input features by pairing approximated Shapley values with multi-level clustering. That work shares this paper’s premise, that a cooperative game is a defensible way to divide credit in a model whose internals resist inspection, but differs on every axis that matters here: the players are input features rather than trained source models, the payoff is cluster quality rather than ranking utility, and the values are approximated rather than enumerated exactly. The present paper moves the unit of analysis from features to architectural components, which is what makes exact enumeration affordable and what makes the retirement question meaningful in the first place: one can switch off a recommender source, but not a feature of a clustering.

Shapley for ranking A recent line develops Shapley attribution specifically for ranked outputs. RankSHAP [17] axiomatises feature attribution for learning-to-rank [18] and evaluates against ranking metrics; RankingSHAP [19] gives a listwise formulation; ShaRP [20] defines games for score, rank, top- k , and pairwise-preference outcomes. Within recommendation, triplet importance [21] values training triplets by ranking utility. All of these take *features* or *training data* as players and approximate the value by sampling. SignalShap differs on both axes: the players are architectural source models, and with five players the value is computed by exhaustive enumeration rather than estimated. The distinguishing design choice is the coalition-independent

candidate set with a coalition-refitted fusion head, which makes coalition values comparable across a subset lattice in a two-stage retrieval-then-rank architecture, a construction these works do not require because their players do not affect retrieval.

Which Shapley, and which game Shapley uniqueness holds *once a game is fixed*; it does not select the game. Sundararajan and Najmi [22] show that plausible operationalisations differ materially, which is why we state our candidate construction, coalition head, and baseline explicitly and treat them as declared modelling choices rather than as the canonical formulation. Alternative cooperative values (Banzhaf and other semivalues [23]) and interaction-aware extensions such as the Shapley interaction index [24] and the Shapley–Taylor index [25] are natural comparators, as are faithful [26] and regression-based [27] interaction formulations. Because every game we fit is non-monotone, recent work on negative interactions in non-convex games [28] is also directly relevant. Because redundancy between sources is central to our setting we evaluate Banzhaf, two binomial semivalues, and the Grabisch–Roubens interaction index rather than deferring them (Section 4.2); we do not compute Shapley–Taylor, and Section 4.2 explains why the distinction matters. Replication sensitivity of Shapley allocations [29] is directly relevant to our duplicate-injection design and is discussed in Section 4.1.

Offline recommender evaluation and exposure bias Logged recommendation data reflect both user choice and earlier exposure policies. Counterfactual risk minimisation addresses learning from logged bandit feedback [30], while propensity-based recommendation evaluation treats recommendations as interventions and corrects selection bias [31]. SignalShap does neither: it attributes an offline metric under the observed-data protocol. The temporal and exposure limitations of that choice are measured in Section 3.4 and

bounded in Section 5.1; no causal or off-policy claim is made.

Hybrid recommenders and ablation practice Burke's taxonomy [32] classifies hybridisation strategies; weighted and feature-combination hybrids dominate practice. Evaluation of component worth is predominantly by ablation [33]. Recent recommender XAI work [34] explains *recommendations*, not *architectures*.

2.2. Background

We collect the definitions the rest of the paper assumes. Readers familiar with cooperative games may skip to Section 3.1.

Cooperative games and the Shapley value A transferable-utility game is a pair $(\mathcal{G}, v)$ with a finite player set $\mathcal{G}$, $n = |\mathcal{G}|$, and a characteristic function $v : 2^{\mathcal{G}} \rightarrow \mathbb{R}$ giving the value each coalition achieves on its own. The Shapley value [1] allocates $v(\mathcal{G})$ among the players by averaging each player's marginal contribution over all orderings, and is the unique allocation satisfying efficiency, symmetry, null player, and additivity. Nothing in the axioms selects v : uniqueness is conditional on a game the modeller declares, which is why Section 3.1 states ours explicitly and Section 4.7 compares alternatives.

For completeness, efficiency requires $\sum_g \phi_g = v(\mathcal{G}) - v(\emptyset)$; symmetry assigns equal values to players with identical marginal contributions; the null-player axiom assigns zero to a player whose marginal contribution is always zero; and additivity requires $\phi(v_1 + v_2) = \phi(v_1) + \phi(v_2)$.

Semivalues Relaxing efficiency yields a family of *semivalues*, each averaging marginal contributions under a pre-declared distribution p_k over the size k of the predecessor coalition:

$$\psi_g(v) = \sum_{k=0}^{n-1} \frac{p_k}{\binom{n-1}{k}} \sum_{\substack{S \subseteq \mathcal{G} \setminus \{g\} \\ |S|=k}} [v(S \cup \{g\}) - v(S)], \quad (1)$$

$$\sum_{k=0}^{n-1} p_k = 1.$$

Shapley is the member with $p_k = 1/n$ for every k : its permutation weight $|S|!(n - |S| - 1)!/n! = 1/\binom{n-1}{|S|}$ places mass $1/n$ on each size and spreads it evenly over the $\binom{n-1}{k}$ coalitions of that size. A "size-uniform semivalue" is therefore not an alternative to Shapley; it is Shapley. Banzhaf is the member with $p_k = \binom{n-1}{k} 2^{-(n-1)}$, which weights all 2^{n-1} coalitions equally and does not satisfy efficiency. The genuinely distinct comparators we use are the binomial semivalues $p_k = \binom{n-1}{k} q^k (1-q)^{n-1-k}$ at $q = 0.25$ and $q = 0.75$, which tilt the weight towards small or large predecessor coalitions respectively. Equation (1) is the master form; Banzhaf, Shapley and the binomial family are the three choices of p_k named above. We report these in Section 4.2.

Interaction indices Pairwise indices measure whether two players are complementary or redundant beyond their main effects, via the second-order discrete derivative $v(S \cup \{a, b\}) - v(S \cup \{a\}) - v(S \cup \{b\}) + v(S)$. They differ in how these terms are weighted: the Shapley interaction index of Grabisch and Roubens [24] and the Shapley–Taylor index [25] use different coefficients and are *not* interchangeable. We use the former throughout.

Leave-one-out, in two distinct forms Leave-one-out ablation reports the single marginal contribution at the grand coalition, $v(\mathcal{G}) - v(\mathcal{G} \setminus \{g\})$. It is a valid estimand for the effect of removing g while everything else remains, and it is not an approximation to the Shapley value; it is one of the 2^{n-1} terms the Shapley value averages.

Which quantity it is computed on matters, and conflating the two is easy. We distinguish the *raw utilities* from the baseline-centred games built on them. Let $U_{\text{rank}}(S)$ and $U_{\text{e2e}}(S)$ be the mean NDCG@10 of coalition S under the fixed-candidate and own-candidate designs respectively, with no baseline subtracted, and let $v_{\text{rank}}(S) = U_{\text{rank}}(S) - b$ and $v_{\text{e2e}}(S) = U_{\text{e2e}}(S) - b(S)$ be the corresponding games. Then

$$\begin{aligned} \text{LOO}_{\text{rank}}(g) &= U_{\text{rank}}(\mathcal{G}) - U_{\text{rank}}(\mathcal{G} \setminus \{g\}), \\ \text{LOO}_{\text{e2e}}(g) &= U_{\text{e2e}}(\mathcal{G}) - U_{\text{e2e}}(\mathcal{G} \setminus \{g\}). \end{aligned} \quad (2)$$

Both are defined on the *raw* utilities, and we flag why. In the fixed-candidate game the baseline b is a single number shared by all 2^n coalitions, so it cancels in the difference and $\text{LOO}_{\text{rank}}(g) = v_{\text{rank}}(\mathcal{G}) - v_{\text{rank}}(\mathcal{G} \setminus \{g\})$ as well. That cancellation *fails* end to end: each coalition retrieves its own $C_u(S)$, so $|C_u|$ and hence $b(S)$ move when a source is retired, and differencing v_{e2e} would add a term $b(\mathcal{G} \setminus \{g\}) - b(\mathcal{G})$ that has nothing to do with the observed change in ranking quality. Defining LOO_{e2e} on the centred game would therefore not give the raw removal cost, and Section 4.7 quantifies the difference on all three corpora.

LOO_{rank} removes g from the *fusion head only*: the candidate set is still the one all five sources retrieved, so g keeps its retrieval role. LOO_{e2e} removes g from retrieval as well, rebuilding C_u from the survivors. The second is what actually happens when a source is switched off in production, and so LOO_{e2e} *equals* the observed single-source retirement loss by construction: it is a definition, not a finding. LOO_{rank} is a different, cheaper quantity that requires no re-retrieval. The empirical question of Section 4.7 is therefore whether the cheap ranking-stage quantity predicts the expensive end-to-end one, and whether it does so better than the ranking-stage Shapley value computed on the same game.

Ranking-stage versus end-to-end games A production recommender retrieves candidates and then ranks them. A *ranking-stage* game holds retrieval fixed and varies only the sources available to the ranker, so coalition values are comparable. An *end-to-end* game lets each coalition retrieve its own candidates, which is closer to a deployment change but makes $v(S)$ depend on two mechanisms at once. We define the former and measure both (Section 4.7).

NDCG with one relevant item Our protocol holds out a single interaction per user, so for a ranking π_u with the held-out item at position r , the normalised discounted cumulative gain [35] reduces to a single term: $\text{NDCG}@10(\pi_u; y_u) = 1/\log_2(r+1)$ if $r \leq 10$ and 0 otherwise, the ideal DCG being 1. The metric is therefore a step function of rank, which is why coalition values move in discrete jumps and why score ties require a deterministic total order.

3. Methodology

3.1. The cooperative game

Let $\mathcal{G} = \{cf, ct, pop, rec, seq\}$ with $|\mathcal{G}| = 5$; Table 1 collects the notation used throughout.

What each source computes Naming the five players and listing their hyperparameters is not enough to reimplement them, so we give the scoring function of each. Write $R \in \{0, 1\}^{m \times |I|}$ for the binarised training matrix, $\mathcal{H}_u$ for user u 's training history, and $t_{\max}$ for the latest training timestamp. All five are fitted on the training fold only and produce a dense score $s_g(u, i)$; every score is z-normalised per user over C_u before it enters the game. That removes each source's location and positive linear scale, so the game is invariant to positive affine rescaling of any source; it does *not* make only the ordering matter. The head fuses the standardised columns linearly, so the relative spacing of a source's standardised scores still affects the fused ranking, and a nonlinear monotone transform of one source can change the result while preserving that source's own ordering.

- **cf, implicit-feedback ALS.** Alternating ridge solves for user and item factors $x_u, y_i \in \mathbb{R}^{64}$ with confidence $c_{ui} = 1 + \alpha r_{ui}$ at $\alpha = 1$, so observed entries carry weight 2 and unobserved entries weight 1. Each half-step solves $(Y^\top Y + \sum_{i \in \mathcal{H}_u} y_i y_i^\top + \lambda_{\text{ALS}} I) x_u = 2 \sum_{i \in \mathcal{H}_u} y_i$ with $\lambda_{\text{ALS}} = 0.1$, for 15 iterations. Score $s_{cf}(u, i) = x_u^\top y_i$.
- **ct, TF-IDF content similarity.** Let $M \in \mathbb{R}^{|I| \times V}$ hold L2-normalised TF-IDF rows over item text ($V = 5000$, tokens split on whitespace). The user profile is the mean of consumed rows, renormalised: $p_u = (|\mathcal{H}_u|^{-1} \sum_{i \in \mathcal{H}_u} M_i) / \|\cdot\|_2$. Score $s_{ct}(u, i) = p_u^\top M_i$, a cosine similarity. If $\mathcal{H}_u$ is empty or the mean vector has zero norm, we set $p_u = \mathbf{0}$. The score is user-specific but time-invariant.
- **pop, time-decayed global popularity.** With half-life $h = 180$ days, each training event of age a days contributes $2^{-a/h}$, where age is measured from $t_{\max}$: $s_{pop}(u, i) = \log(1 + \sum_{(u', i, t) \in \mathcal{D}_{\text{tr}}} 2^{-(t_{\max} - t)/(86400h)})$. The score varies across items but not across users, so every user sees the same global popularity ordering. It is *not* constant along a candidate row, which is why it survives z-normalisation and can carry a standalone value.

Table 1

Core notation.

Symbol	Meaning
$\mathcal{G}$	five source players
$\mathcal{U}, \mathcal{U}_{\text{eval}}$	retained users and users with nonempty candidates
$C_u, C_u(S)$	fixed and coalition-specific candidate sets
$Z_u^{(S)}, w^{(S)}$	standardized source scores and coalition head
b_u	expected NDCG@10 of a random ranking of C_u
$v_u(S), v(S)$	per-user and mean baseline-centred utilities

- **rec, recency over content clusters.** Items are assigned to one of 20 clusters by argmax over a 20-component truncated SVD of the item TF-IDF matrix ($V = 2000$). For cluster $\kappa(i)$, let $\ell_{u\kappa}$ be the timestamp of u 's most recent training interaction in that cluster. Then $s_{rec}(u, i) = 2^{-(t_{\max} - \ell_{u\kappa(i)})/(86400 \cdot 30)}$ if $\ell_{u\kappa(i)}$ exists and 0 otherwise. It shares its input with ct by construction, which is the pre-declared overlap of Section 3.4. Replacing global $t_{\max}$ by a user's scoring time multiplies that user's nonzero recency scores by one positive constant, so the subsequent per-user z-normalisation removes the choice whenever the row is nonconstant. The argmax partition is not invariant to an arbitrary sign reversal of an SVD component, so the implementation canonicalises each component to have a positive largest-magnitude loading before taking the argmax, which makes the partition a function of the data alone; rec is then bit-identical across solver seeds, which we test. We note for exactness that this convention is also what scikit-learn already applies internally, so the signs were never in fact the operative hazard, and imposing it explicitly changes no reported number. What sign canonicalisation cannot repair is *degeneracy*: where two singular values coincide, the corresponding subspace has no preferred basis, different solvers return different rotations of it, and the argmax partition genuinely moves. That is the residual reproducibility caveat for rec , and it is a property of the spectrum rather than of the solver seed.
- **seq, PPMI-SVD sequential embedding.** Within each user's time-ordered history, all item pairs at distance ≤ 5 contribute to a symmetric co-occurrence count C . We set $\text{PPMI}_{ij} = 0$ when $C_{ij} = 0$ and otherwise use $\text{PPMI}_{ij} = \max\{0, \log[C_{ij}(\sum_{ab} C_{ab}) / (C_i \cdot C_j)]\}$. A 64-component truncated SVD gives L2-normalised item embeddings e_i . The user vector is a recency-weighted mean of the last 5 items, weights 0.8^k for the k -th most recent, and $s_{seq}(u, i) = \tilde{u}^\top e_i$.

The ct - rec pair is the declared content overlap and the cf - pop pair the declared popularity overlap; both are stated before any attribution is computed.

Coalition-independent candidates For user u and source g , let $T_g(u, N_g)$ be the first N_g eligible items under the total

order $(-s_g(u, i), \text{item index})$: higher score first, with item index breaking every score tie, including ties at the top- N_g boundary. Set

$$C_u = \text{Trunc}_{N_{\max}^{(d)}}\left(\bigcup_{g \in \mathcal{G}} T_g(u, N_g)\right), \quad |C_u| \leq N_{\max}^{(d)}. \quad (3)$$

The implementation must select from this full lexicographic source order; a partial selection routine that chooses an arbitrary subset of cutoff ties does not implement the definition. Here d indexes the dataset, since the cap is frozen per corpus (Table 2), and Trunc_N retains the first N items after sorting the union by the key $(-\sum_{g \in \mathcal{G}} 1/\text{rank}_g(i), \text{item index})$, where $\text{rank}_g(i) \in \{1, \dots, N_g\}$ is the position of i in source g 's own top- N_g list. The sum runs only over the sources that retrieved i ; a source that did not contributes 0, equivalently $1/\text{rank}_g(i) \rightarrow 0$ as an unretrieved item's rank passes N_g . Both components of the truncation key are invariant under permuting $\mathcal{G}$, so relabelling the players cannot change the game. The construction contains no random operation of its own and is therefore deterministic *conditional on the fitted source models*; those models are trained with the experiment seed (Table 3), so C_u can still vary across seeds. The bound in (3) holds by construction.

Sources begin at equal depth $N_g = \lceil N_{\max}^{(d)}/|\mathcal{G}| \rceil$. If the union is smaller than the cap, all depths are increased by $\lceil (N_{\max}^{(d)} - |C_u|)/|\mathcal{G}| \rceil$ and the lists are rebuilt, for at most ten passes or until every eligible source list is exhausted. If fewer than $N_{\max}^{(d)}$ distinct eligible items exist, all of them are retained. For an end-to-end coalition S , the same procedure uses $|S|$ in place of $|\mathcal{G}|$; the empty coalition is handled by convention and retrieves no list.

Let $\mathcal{U}$ be the users retained after preprocessing and define

$$\mathcal{U}_{\text{eval}} = \{u \in \mathcal{U} : |C_u| > 0\}. \quad (4)$$

Users outside $\mathcal{U}_{\text{eval}}$ are excluded from every coalition mean and coalition-head fit and are reported separately as candidate-construction failures. Candidate recall uses the same denominator, $|\mathcal{U}_{\text{eval}}|^{-1} \sum_{u \in \mathcal{U}_{\text{eval}}} \mathbb{1}[i_u^{\text{test}} \in C_u]$, while candidate coverage is $|\mathcal{U}_{\text{eval}}|/|\mathcal{U}|$.

For $u \in \mathcal{U}_{\text{eval}}$, source scores enter the fusion head through

$$z_{u,g,i} = \begin{cases} (s_g(u, i) - \mu_{u,g})/\sigma_{u,g}, & \sigma_{u,g} > 0, \\ 0, & \sigma_{u,g} = 0, \end{cases} \quad (5)$$

$$\mu_{u,g} = \text{mean}_{i \in C_u} s_g(u, i),$$

where $\sigma_{u,g}$ is the corresponding standard deviation. A source is structurally nondegenerate when $\sigma_{u,g} > 0$ for at least one evaluated user; this is the criterion used by the pre-run audit.

Source-order invariance The audited legacy truncation key used source order as a tie-break, so relabelling the players could alter C_u . Under all 120 source permutations on a MovieLens probe, the worst mean Jaccard overlap with the original sets was 0.9966. Replacing that key with

the reciprocal-rank sum above changed the paired within-platform Shapley vector by at most 2.8×10^{-5} , preserved its ordering and material cf sign disagreement, and left recall at 0.748. Cross-platform BLAS differences are larger than this key effect, so all such contrasts are made within one platform. The ten-seed attribution and retirement results use the symmetric rule. Single-seed diagnostics are classified by candidate-rule provenance in Section 4.6.

This choice is load-bearing. The obvious alternative, drawing one candidate pool from the full fused scorer, is *not aligned with the ranking-stage estimand we declare*, because it hands the grand coalition a retrieval pool selected in its own favour: every $v(S)$ for $S \subsetneq \mathcal{G}$ is then evaluated on items chosen to suit $\mathcal{G}$. We state this as a construct-design argument, not a theorem. Nothing here proves that a $\mathcal{G}$ -selected pool must raise $v(\mathcal{G})$ for an arbitrary relevance distribution; what it does is condition the whole lattice on a pool chosen by one distinguished coalition, so the differences $v(S \cup \{g\}) - v(S)$ that Shapley aggregates no longer isolate fusion. Under the union design each coalition scores an identical item set, so those differences reflect *fusion within* S , not retrieval.

We retain the rejected design as an ablation and measure the effect rather than assert its direction. On MovieLens-1M (seed 42) the grand-coalition pool raises candidate recall from 0.748 to 0.807 and $v(\mathcal{G})$ from 0.05136 to 0.05547, a difference of +0.00411, and it changes φ_{ct} from -0.00009 to +0.00152, the one source whose sign it flips. On this corpus the direction is the one the design argument anticipates, which supports the choice without establishing it in general.

Candidate recall is a ceiling, not a detail A test item outside C_u contributes zero to NDCG@10 by construction. Candidate recall $\Pr[\text{test}_u \in C_u]$ therefore upper-bounds every ranking metric downstream, and we report it before any attribution number (Table 2).

Characteristic function Let $z_{u,g,i}$ be the per-user, per-source z -normalised score. For coalition S , $\hat{s}_{u,i}(S) = \sum_{g \in S} w_g^{(S)} z_{u,g,i}$ with $w^{(S)}$ a head fitted on the validation fold. Let $y_{u,i} = \mathbb{1}[i = i_u^{\text{test}}]$ for $i \in C_u$ be the binary relevance vector. For $S \neq \emptyset$, let $\pi_u(S)$ be the ranking of C_u under the total order $(-\hat{s}_{u,i}(S), \text{item index})$; no approximate-equality tolerance is used. With cutoff $K = 10$, define the per-user raw utility, baseline-centred game, and aggregate game piecewise:

$$b_u = \mathbb{E}_{\pi \sim \mathcal{U}(C_u)} [\text{NDCG}@10(\pi; y_u)],$$

$$U_u(S) = \begin{cases} b_u, & S = \emptyset, \\ \text{NDCG}@10(\pi_u(S); y_u), & S \neq \emptyset, \end{cases}$$

$$v_u(S) = U_u(S) - b_u,$$

$$v(S) = \frac{1}{|\mathcal{U}_{\text{eval}}|} \sum_{u \in \mathcal{U}_{\text{eval}}} v_u(S). \quad (6)$$

The baseline is the *expected* payoff of a uniformly random ranking, not a sampled one. With a single held-out item and

IDCG = 1 it is available in closed form: the item is equally likely to land in any of the $|C_u|$ positions, so

$$b_u = \frac{\mathbb{1}[i_u^{\text{test}} \in C_u]}{|C_u|} \sum_{r=1}^{\min(K, |C_u|)} \frac{1}{\log_2(r+1)}, \quad u \in \mathcal{U}_{\text{eval}}. \quad (7)$$

If $i_u^{\text{test}} \notin C_u$ then $y_u = \mathbf{0}$ and $\text{NDCG@10}(\cdot; y_u) = 0$ for every ranking, so $v_u(S) = 0$; such users contribute zero to every coalition and are the reason candidate recall bounds all downstream metrics. The convention $U_u(\emptyset) = b_u$ makes $v_u(\emptyset) = 0$ per user, rather than only on average. The empty coalition therefore has no deterministic item-index ranking; its utility is the declared expected-random reference value.

The coalition head is the ridge solution on the validation fold,

$$w^{(S)} = \arg \min_{w \in \mathbb{R}^{|S|}} \sum_{u \in \mathcal{U}_{\text{eval}}} \frac{1}{|C_u|} \left\| \mathbf{y}_u^{\text{val}} - Z_u^{(S)} w \right\|_2^2 + \lambda \|w\|_2^2, \quad (8)$$

where $Z_u^{(S)} \in \mathbb{R}^{|C_u| \times |S|}$ stacks the z-normalised scores of the sources in S over $i \in C_u$, and $\mathbf{y}_u^{\text{val}} \in \{0, 1\}^{|C_u|}$ marks the user's held-out validation interaction. The $1/|C_u|$ factor matters: candidate sets differ in size across users, and without it users with larger pools would dominate the fit. Weights are unconstrained and no intercept is fitted. Test interactions are never used to fit $w^{(S)}$. The objective is a *sum* over users, not an average, matching the implementation that generated the present results. Consequently the numerical penalty is sample-size dependent: under an averaged loss its effective value would be $\lambda/|\mathcal{U}_{\text{eval}}|$. The frozen $\lambda = 1$ is therefore a within-corpus specification, not an equal regularisation strength across corpora. We consequently make no cross-corpus claim about the effect of a common numerical λ ; an averaged-loss rerun with retuning is required for such a comparison.

Why the baseline is an expectation and not a draw The baseline cancels in $v_u(S \cup \{g\}) - v_u(S)$ whenever $S \neq \emptyset$, but not in the empty-to-singleton marginal. Since the empty set carries Shapley weight $w(0) = 1/n$, changing the baseline shifts every ϕ_g by exactly $-\Delta/n$, where Δ is the change in the mean baseline. A diagnostic draw that moved the mean baseline by -0.000796 shifted all five MovieLens values by $+0.000796/5$ and changed the sign of a near-zero source. Equation (7) removes that arbitrary draw; efficiency is unchanged because $v(\mathcal{G})$ shifts with the same baseline.

A residual, quantified source of cross-platform variation Apple-silicon and x86-64 runs with identical configuration differ in candidate contents for 2 of 6035 MovieLens users because BLAS accumulation changes near-ties at the truncation boundary. This moves $v(\mathcal{G})$ by approximately 5×10^{-4} and the raw monotonicity count, while preserving the material count and all reported signs. “Exact” therefore describes aggregation over a given fitted game,

not bit-identical training across numerical backends; the reproducibility record must therefore include platform and library versions.

The regularisation strength is frozen λ is chosen once on a pilot and never tuned per coalition. This is a comparability choice, and we state it as such rather than as a proved bias result. Tuning λ inside each coalition would make every $v(S)$ the value of a differently specified head, so a difference $v(S \cup \{g\}) - v(S)$ would mix the contribution of g with a change of model-selection procedure. Larger coalitions also have more parameters and more scope to fit the held-out interaction, so the selection effect need not be constant in $|S|$; we do not claim a theorem that it must grow with $|S|$ or that it must raise $v(\mathcal{G})$, only that a frozen λ removes the confound. Section 4.3 reports the attributions across $\lambda \in [10^{-5}, 10^3]$ so the reader can see the sensitivity directly. A properly nested tuning scheme, with an inner fold per coalition, is a legitimate alternative design and we do not evaluate it here.

Exact aggregation

$$\phi_g = \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|! (|\mathcal{G}| - |S| - 1)!}{|\mathcal{G}|!} [v(S \cup \{g\}) - v(S)], \quad (9)$$

an exact sum over all $2^5 = 32$ coalitions; for $n = 5$ the weights by $|S|$ are $(\frac{1}{5}, \frac{1}{20}, \frac{1}{30}, \frac{1}{20}, \frac{1}{5})$.

What “exact” means The aggregation has *no sampling error*, unlike Monte-Carlo Shapley. But $v(S)$ is itself *fitted*, so ϕ_g carries estimation error. We write “exact given the fitted v ” throughout, and report seed-based run-to-run intervals in the main text rather than an appendix. Efficiency, by contrast, holds exactly regardless of fit noise because it is structural.

3.2. Algorithm and cost

Algorithm 1 states the procedure end to end. Two details carry the correctness of everything downstream and are easy to get wrong. The candidate set is built *once*, from all sources, and reused unchanged for every coalition: building it per coalition would make $v(S)$ a different estimand: a coalition would be scored on candidates it selected for itself, so coalition differences would combine retrieval reach with ranking quality. That game is perfectly well defined, and Section 4.7 evaluates it as v_{e2e} ; we fix the candidate set here because the present estimand is ranking-stage attribution under a common retrieval ceiling. And the ridge head for each coalition is fit against *validation* targets while $v(S)$ is scored against *test* items, so no coalition value is measured in-sample.

The temporal state is frozen at the training fold, deliberately The history the test ranking is produced from deserves an explicit statement, because the split is *leave-last-two-out*. Every source is fitted on the training fold alone, and the score matrices are masked against training items alone.

The validation interaction i_u^{val} , chronologically the second-to-last event, is therefore *not* appended to the user's history before test scoring and *not* masked out of the test ranking. One frozen source state serves both folds. This is a two-step-ahead evaluation from a frozen state, not a one-step-ahead evaluation, and we name it as such rather than let "leave-last-out" imply otherwise.

The choice is deliberate: refreshing the state between folds would make the head-fitting features and the evaluation features come from two different scorers, so a coalition difference $v(S \cup \{g\}) - v(S)$ would mix the contribution of g with a change of source state. It is not free. One consequence is directional and one is not, and they are easily conflated.

Holding the source scores and C_u fixed, and conditional on $i_u^{\text{val}} \neq i_u^{\text{test}}$, masking the consumed validation item can only preserve or improve the test item's rank. Event-level repeats are handled explicitly rather than dropped. If validation and test refer to the same item and that item is unseen in training, it may be both the validation positive and the test positive. If the test item already appears in training, the training-history mask makes it ineligible, so its test target is the zero vector and it contributes a candidate miss. These cases remain in $\mathcal{U}_{\text{eval}}$ when C_u is nonempty. We count per corpus the users whose validation and test events name the same item, the users whose test item already appears in training, and the corpus-level repeat-event rate. We measured rather than assumed, and the counts are not negligible (Table 4).

MovieLens-1M is deduplicated at source, so both cases are exactly absent and the frozen-state argument applies to every user unconditionally. Amazon-VG carries a small rate of both. Gowalla is the corpus the concern was raised about, and it is worse than anticipated: over half of all check-in events repeat a venue the same user already visited, 14.21% of users have the same venue as validation and test target, and *half of all evaluated users have a test venue that already appears in their training history*.

The last column is the consequential one, and it is not a small correction. For such a user the training-history mask sets the test item's score to $-\infty$ in every source, so it is retrieved into no candidate set, $\text{NDCG}_u = 0$ under every coalition, and the baseline $b_u = 0$ as well. Hence $v_u(S) = 0$ identically, for all 32 coalitions. Such users are not noise in the game; they are exact zeros. Because $v(S)$ is a mean over $\mathcal{U}_{\text{eval}}$, they act as a pure multiplicative dilution: writing ρ for the fraction of evaluated users whose test item is retrievable at all,

$$v(S) = \rho \cdot v^{\text{live}}(S) \quad \text{for every } S \subseteq \mathcal{G}, \quad (10)$$

where v^{live} is the same game restricted to those users. Shapley is linear in v , so $\phi_g = \rho \phi_g^{\text{live}}$ for every source, and likewise for LOO and for every LOO-versus-Shapley gap reported in Section 4.7. We verified Eq. (10) numerically: over all 32 coalitions and all five attributions the two sides agree to 1.4×10^{-17} , machine precision.

Two consequences follow, and they point in opposite directions. The reassuring one is that every *ordering*, every

sign, every Kendall τ , and every ratio-to- $v(\mathcal{G})$ this paper reports on Gowalla is exactly invariant to the dilution, because a positive common factor cancels in all of them. The material *pop* flip is 12.2% of $v(\mathcal{G})$ whether or not the masked users are counted. No qualitative claim changes. The uncomfortable one is that Gowalla's absolute magnitudes are, on the frozen protocol, roughly half of what the same game measures on the users it can actually score: $\rho = 0.501$, so $v(\mathcal{G}) = 0.01705$ corresponds to $v^{\text{live}}(\mathcal{G}) = 0.03403$ and the *pop* gap of 0.00207 to 0.00413. We report the diluted numbers throughout, because $\mathcal{U}_{\text{eval}}$ is the declared evaluation population and rescaling after seeing ρ would be selection, but absolute Gowalla values should be read as a property of that population and not compared directly against the other two corpora, where ρ is 1.000 and 0.972.

The same factor explains part of Gowalla's recall shortfall. Candidate recall cannot exceed ρ by construction, so the 0.60 gate was never reachable on this corpus under the training-history mask, whatever N_{max} is chosen: the measured 0.450 is 0.898 *conditional on the test item being retrievable at all*, and validation recall 0.462 is likewise 0.922 conditional. Section 3.5 presents the gate discussion with this ceiling made explicit. The shortfall is therefore not attributable entirely to catalogue size and candidate-pool affordability, as a sparsity-only reading would have it: roughly half of it is the repeat-visit structure of check-in data interacting with a mask designed for corpora without repeats.

A refreshed one-step-ahead protocol also changes sequential and recency scores, candidate retrieval, and the baseline, so frozen and refreshed games are not ordered in general. We measured the net effect with the head fitted in the validation state and the test coalition scored after appending and masking the validation event. Refreshing raises $v(\mathcal{G})$ from 0.05169 to 0.07026 on MovieLens-1M (+36%), from 0.04130 to 0.04902 on Amazon-VG (+19%), and from 0.01690 to 0.01753 on Gowalla (+3.7%). Candidate recall rises from 0.748 to 0.779 and from 0.589 to 0.604 on the two denser corpora, but falls from 0.4504 to 0.4488 on Gowalla, confirming that no general monotonicity claim is available.

The source ordering is unchanged on MovieLens and Gowalla ($\tau = 1.00$). On Amazon-VG it changes ($\tau = 0.80$), and the leading source switches from *cf* to *seq*. Absolute values and the Amazon-VG leader are therefore protocol-dependent; the allocation-versus-removal comparison is the narrower robust target.

Validation positives outside the candidate set The head is fitted on $\mathbf{y}_u^{\text{val}}$, the indicator of i_u^{val} within C_u . When the validation item is not retrieved, that vector is all zero and the user contributes negatives only. We record validation candidate recall alongside test recall so the size of this effect becomes visible rather than implicit, and we retain the all-zero users: dropping them would fit the head on a subpopulation selected for being easy to retrieve, which is a worse bias than the one it removes.

It is tempting to read such a user as merely shrinking the fitted weights without distorting their direction. That is false. Such a user contributes its positive-semidefinite block to A but nothing to c , and for $w = (A + \lambda I)^{-1}c$ with c unchanged, adding a PSD term to A rescales the solution unevenly across eigendirections, so it can rotate w as well as shorten it; a two-source numerical example moves the weight vector by 12.8° while shrinking its norm. The correct statement is that validation misses add regularisation in the directions of that user's source-score covariance, changing both the magnitude and the orientation of the coefficient vector. We therefore treat the retention policy as a declared choice whose sensitivity should be tested, not as a harmless one, and we report validation recall so that a reader can judge how many users it concerns: 0.799 on MovieLens-1M (1 216 of 6 035 fitted users contribute negatives only), 0.648 on Amazon-VG (2 505 of 7 120), and 0.462 on Gowalla, where 4 773 of 8 865 users, a clear majority, fit on negatives alone. The Gowalla figure is largely the same mechanism as the test-side dilution: 50.74% of its users have a validation venue already present in training (Table 4), which the mask makes unretrievable, so their validation target is all zero by construction rather than by retrieval failure. Those are not marginal fractions, so the policy needed testing rather than defending. The sensitivity analysis refits with those users dropped. The attributions barely move: $\max_g |\Delta \varphi_g| = 3.7 \times 10^{-5}$ on MovieLens and 1.8×10^{-4} on Gowalla, both with identical orderings, and 2.1×10^{-4} on Amazon-VG, where the ordering shifts only among *ct* and *pop*, separated by 3×10^{-4} . Every sign agrees under both policies on all three corpora. That the sparsest corpus, with the most affected users, is also among the least sensitive is the clearest evidence that the retention choice is not load-bearing. The retention choice is therefore immaterial here, which we can now say from measurement rather than from an argument about bias.

Cost Let $n = |\mathcal{G}|$, m users, $N = |C_u|$, and d the item-embedding dimension. Source training is the usual cost of the five components and is independent of the game. Candidate construction scores every eligible item under every source, so the retrieval stage carries a C_{score} term that depends on the catalogue $|I|$, not on N . Writing C_{score} for the cost of scoring all $m|I|$ user-item pairs across the n sources, top- N selection by partial sort adds $O(mn|I|)$ and ordering the union of up to nN retrieved entries adds $O(mnN \log(nN))$. For the *game*, accumulating the Gram matrix once costs $O(mNn^2)$, after which each of the 2^n coalitions is an $O(n^3)$ solve on a slice, $O(mNn)$ to form the fused scores in the worst case, and $O(mN \log N)$ to rank and score. Total is

$$O(C_{\text{score}} + mn|I| + mnN \log(nN) + mNn^2 + 2^n(n^3 + mNn + mN \log N)). \quad (11)$$

The naive implementation rebuilds the design matrix per coalition, giving $O(2^n \cdot mNn^2)$; on MovieLens that difference was 81 minutes against 40 seconds. Memory is dominated

Algorithm 1 Exact ranking-stage SignalShap

Require: train $\mathcal{D}_{\text{tr}}$, validation $\mathcal{D}_{\text{val}}$, test $\mathcal{D}_{\text{te}}$; sources $\mathcal{G}$, $n = |\mathcal{G}|$; cap N_{max} ; growth limit $L = 10$; penalty λ ; cutoff K

Ensure: coalition values $v(S)$, source values φ_g , per-user $\varphi_g(u)$

- 1: fit every source $g \in \mathcal{G}$ on $\mathcal{D}_{\text{tr}}$ only
- 2: score all eligible items per user from the TRAINING fold only; mask items seen in training $\triangleright$ frozen-state protocol, see below
- 3: **for** each user u **do**
- 4: order each source list by descending score, then item index
- 5: initialise $N_g \leftarrow \lceil N_{\text{max}}/n \rceil$ for every source
- 6: grow $\bigcup_g T_g(u, N_g)$ for at most L passes as in Eq. (3); truncate by reciprocal-rank sum, then item index
- 7: **if** $|C_u| = 0$ **then**
- 8: record an empty-candidate diagnostic and continue
- 9: **end if**
- 10: add u to $\mathcal{U}_{\text{eval}}$
- 11: $Z_u \in \mathbb{R}^{|C_u| \times n} \leftarrow$ scores standardised by Eq. (5) $\triangleright$ constant source rows become zero
- 12: $\mathbf{y}_u^{\text{val}}, \mathbf{y}_u^{\text{te}} \in \{0, 1\}^{|C_u|} \leftarrow$ indicators of the validation and test items
- 13: $b_u \leftarrow \frac{1}{|C_u|} \sum_{r=1}^{\min(K, |C_u|)} \frac{1}{\log_2(r+1)}$ if the test item lies in C_u , else 0 $\triangleright$ Equation 7; deterministic, no permutation is drawn
- 14: **note** $\mathbf{y}_u^{\text{val}}$ is NOT added to the history and NOT masked before test scoring: one frozen source state serves both folds
- 15: record whether validation equals test and whether the test item was observed in training
- 16: **end for**
- 17: $A \leftarrow \sum_{u \in \mathcal{U}_{\text{eval}}} |C_u|^{-1} Z_u^T Z_u$, $c \leftarrow \sum_{u \in \mathcal{U}_{\text{eval}}} |C_u|^{-1} Z_u^T \mathbf{y}_u^{\text{val}}$ $\triangleright$ VALIDATION targets; accumulate once
- 18: **for** each $S \subseteq \mathcal{G}$ **do**
- 19: **if** $S = \emptyset$ **then**
- 20: $v_u(S) \leftarrow 0$ for all $u \in \mathcal{U}_{\text{eval}}$ $\triangleright v(\emptyset) = 0$ exactly and per user
- 21: **else**
- 22: **if** $\lambda > 0$ **then**
- 23: solve $[A_{SS} + \lambda I_{|S|}] w^{(S)} = c_S$ $\triangleright$ do not form an explicit inverse
- 24: **else**
- 25: $w^{(S)} \leftarrow A_{SS}^+ c_S$ $\triangleright$ minimum-norm; A_{SS} is singular under exact duplicates
- 26: **end if**
- 27: $v_u(S) \leftarrow \text{NDCG@K}(\text{rank}(Z_u^{(S)} w^{(S)}); \mathbf{y}_u^{\text{te}}) - b_u$ for all $u \in \mathcal{U}_{\text{eval}}$ $\triangleright$ descending score, then item index; no tolerance relation
- 28: **end if**
- 29: $v(S) \leftarrow |\mathcal{U}_{\text{eval}}|^{-1} \sum_{u \in \mathcal{U}_{\text{eval}}} v_u(S)$
- 30: **end for**
- 31: **for** each $g \in \mathcal{G}$ **do**
- 32: $\varphi_g \leftarrow \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|!(n-|S|-1)!}{n!} [v(S \cup \{g\}) - v(S)]$
- 33: $\varphi_g(u) \leftarrow \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|!(n-|S|-1)!}{n!} [v_u(S \cup \{g\}) - v_u(S)]$
- 34: **end for**
- 35: verify efficiency $\sum_g \varphi_g = v(\mathcal{G})$, the per-user decomposition, source-order invariance of C_u , and that no player is structurally degenerate according to Eq. (5)

not by the game but by the n dense score matrices, $4mn|I|$ bytes in single precision, or 129 GB at full Gowalla size, which is why the corpora are subsampled to a declared budget.

The 2^n factor is the binding constraint on scaling. At $n = 5$ exhaustive enumeration costs 32 refits and is unambiguously worth it; at $n = 20$ it would be 10^6 and sampling becomes necessary, at which point the paper's central claim of an exact, variance-free aggregation no longer holds. We regard $n \lesssim 12$ as the practical range, which covers the architectural-source setting this paper addresses but not feature-level attribution.

Measured wall-clock for the full E0–E8 pipeline on an Apple M4 with 48 GB, single seed: MovieLens 163 s, Amazon-VG 71 s, Gowalla 943 s, of which the 32 coalition evaluations are 13, 10, and 285 s respectively. The game itself is never the bottleneck; candidate construction and the neural reference baselines dominate.

3.3. Formal properties

Properties 1–3 are instantiations of standard Shapley axioms; we claim no novelty for their mathematical content, only for their careful use here.

Property 1 (Efficiency). $\sum_{g \in \mathcal{G}} \varphi_g = v(\mathcal{G})$.

Property 2 (LOO redundancy collapse). Let v be **monotone**, i.e. $v(S \cup \{g\}) \geq v(S)$ for all S and $g \notin S$. If g_1, g_2 satisfy $v(S \cup \{g_1\}) = v(S \cup \{g_2\}) = v(S \cup \{g_1, g_2\})$ for every $S \subseteq \mathcal{G} \setminus \{g_1, g_2\}$, then $\text{LOO}(g_1) = \text{LOO}(g_2) = 0$ while $\varphi_{g_1} = \varphi_{g_2} \geq v(\{g_1\})/|\mathcal{G}| > 0$ whenever $v(\{g_1\}) > 0$.

Proof. Take $S = \mathcal{G} \setminus \{g_1, g_2\}$ in the redundancy condition. Then $v(\mathcal{G}) = v(\mathcal{G} \setminus \{g_1\}) = v(\mathcal{G} \setminus \{g_2\})$, so both grand-coalition LOO terms are zero. Exchanging g_1 and g_2 leaves every coalition value unchanged, hence Shapley symmetry gives $\varphi_{g_1} = \varphi_{g_2}$. Under monotonicity every marginal contribution in Eq. (9) is nonnegative. Retaining only the $S = \emptyset$ term, whose coefficient is $1/|\mathcal{G}|$, gives $\varphi_{g_1} \geq [v(\{g_1\}) - v(\emptyset)]/|\mathcal{G}| = v(\{g_1\})/|\mathcal{G}|$ because $v(\emptyset) = 0$. $\square$

The monotonicity hypothesis is not optional It is tempting to state Property 2 without it. That statement is **false**. Consider three players with $v(\emptyset) = 0$, $v(g_1) = v(g_2) = 1$, $v(g_3) = 5$, $v(g_1g_2) = 1$, $v(g_1g_3) = v(g_2g_3) = 0.5$, and $v(\mathcal{G}) = 0.5$. Redundancy holds exactly, $v(\{g_1\}) = 1 > 0$, and $\text{LOO}(g_1) = \text{LOO}(g_2) = 0$ as predicted. Symmetry and efficiency both hold. Yet

$$\begin{aligned} \varphi_{g_1} = \varphi_{g_2} &= -\frac{5}{12} \approx -0.4167, \\ \varphi_{g_3} &= \frac{4}{3} \approx 1.3333, \quad \Sigma = 0.5 = v(\mathcal{G}). \end{aligned}$$

The redundant sources receive *negative* credit despite positive standalone value. Restoring monotonicity repairs it: on the monotone variant ($v(g_1g_3) = v(g_2g_3) = v(\mathcal{G}) = 5.5$) one obtains $\varphi_{g_1} = 5/12 \approx 0.4167 \geq v(\{g_1\})/3 = 1/3$.

This matters practically: the coalition-conditional refit is precisely the mechanism that can break monotonicity, since a head fitted on $S \cup \{g\}$ may score worse than one fitted on

S . We therefore *audit* monotonicity on all $|\mathcal{G}| \cdot 2^{|\mathcal{G}|-1} = 80$ pairs rather than assume it, and report violations as findings.

Property 3 (Free per-user decomposition). By linearity of the Shapley operator and the definition of v in Eq. (6),

$$\varphi_g = \frac{1}{|\mathcal{U}_{\text{eval}}|} \sum_{u \in \mathcal{U}_{\text{eval}}} \varphi_g(u). \quad (12)$$

Lemma 1. *Exact redundancy is a measure-zero condition that is unlikely to hold on data without controlled duplication, so Property 2 motivates but cannot instantiate an empirical test. We therefore use a two-sided condition. Call g_1, g_2 ϵ -substitutable if, for every $S \subseteq \mathcal{G} \setminus \{g_1, g_2\}$,*

$$\max \left\{ |v(S \cup \{g_1\}) - v(S \cup \{g_1, g_2\})|, |v(S \cup \{g_2\}) - v(S \cup \{g_1, g_2\})| \right\} \leq \epsilon. \quad (13)$$

Then:

- (i) $|\text{LOO}(g_1)| \leq \epsilon$ and $|\text{LOO}(g_2)| \leq \epsilon$.
- (ii) If in addition v is monotone and $v(\{g_1\}) \geq c$, then $\varphi_{g_1} \geq c/|\mathcal{G}|$, with no ϵ correction.

Proof. (i) Taking $S = \mathcal{G} \setminus \{g_1, g_2\}$ in (13) gives $|v(\mathcal{G} \setminus \{g_2\}) - v(\mathcal{G})| \leq \epsilon$ and $|v(\mathcal{G} \setminus \{g_1\}) - v(\mathcal{G})| \leq \epsilon$, which are exactly $|\text{LOO}(g_2)| \leq \epsilon$ and $|\text{LOO}(g_1)| \leq \epsilon$. The two-sided form of (13) is what makes both statements available; a one-sided condition bounds only one of them.

(ii) Write the Shapley value in permutation form, $\varphi_{g_1} = \sum_{T \subseteq \mathcal{G} \setminus \{g_1\}} w(|T|) [v(T \cup \{g_1\}) - v(T)]$ with $w(t) = t!(n-t-1)!/n!$ and $n = |\mathcal{G}|$. Under monotonicity every summand is non-negative, so discarding all terms except $T = \emptyset$ can only decrease the total:

$$\varphi_{g_1} \geq w(0) [v(\{g_1\}) - v(\emptyset)] = \frac{1}{n} v(\{g_1\}) \geq c/n.$$

Two remarks. First, the bound in (ii) follows from monotonicity alone and does *not* require substitutability; we state the two parts together only because the pair is what makes the LOO–Shapley gap quantitative. Second, the anchor weight is $w(0) = 1/n$, not the mass $\sum_{T \subseteq \mathcal{G} \setminus \{g_1, g_2\}} w(|T|) = 1/2$ that corresponds to $\Pr[g_1 \text{ precedes } g_2]$ under a uniform random ordering. Conflating the two would overstate the bound by a factor of $n/2$. $\square$

Combining the two parts, an ϵ -substitutable pair in a monotone game has a LOO–Shapley gap of at least $c/n - \epsilon$ for each member. This is the quantitative form of the qualitative statement in Property 2.

Sharpness of the bound. The constant in Lemma 1(ii) is $1/|\mathcal{G}|$, *never* $1/2$: substituting the total weight mass for the anchor weight overstates the bound by $n/2$, a factor of 2.5 at $n = 5$, and is unattainable: on the monotone game above, $\frac{1}{2}v(\{g_1\}) = 0.5 > 0.4167 = \varphi_{g_1}$. Relatedly, no pairwise closed form $\frac{1}{2}[v(\mathcal{G}) - v(\mathcal{G} \setminus \{g_1, g_2\})]$ is valid; it returns -2.25 against a true -0.4167 on the counterexample.

Remark 1 (Positive-affine scale invariance). Per-user z -normalisation makes φ_g invariant under user- and source-specific maps $z \mapsto az+b$ with $a > 0$. The restriction matters: a negative a reverses the induced ranking and is not an invariance, and the normalisation is undefined for a constant score vector, which we handle by mapping zero-variance sources to the zero vector. Invariance does not extend to nonlinear monotone maps.

3.4. Experimental setup

Players *cf*: implicit-feedback ALS [36], 64 factors. *ct*: TF-IDF cosine over item metadata. *pop*: time-decayed global frequency. *rec*: exponential recency over an item's content cluster. *seq*: item2vec, realised as SVD of the PPMI co-occurrence matrix. Levy and Goldberg [37] show that skip-gram with negative sampling implicitly factorises a shifted PMI matrix; we factorise it directly, which is deterministic on CPU. We call this a PPMI-SVD sequential model rather than claiming exact algorithmic equivalence to skip-gram.

Expected overlap, stated in advance Two pairs overlap by construction and we state this before reporting results rather than present it as a discovery: *pop-cf* (ALS on implicit feedback chases popularity) and *rec-ct* (recency is defined over content clusters). RQ2 asks whether the fitted game reflects this predeclared qualitative expectation; it is a diagnostic rather than ground-truth validation of the allocation.

Baselines and comparators The attribution comparators are ranking-stage LOO, a fixed-head masking game, an end-to-end coalition-retrieval game, Banzhaf values, and two binomial semivalues. The retirement experiment compares Shapley and ranking-stage LOO against observed end-to-end removal losses. Appendix A uses a globally tuned head of the same family as its matched baseline, with uniform and popularity fusion as heuristic references. LightGCN and SASRec are shown only as contextual recommendation references because their tuning budgets are not matched; they support no superiority claim.

Corpora Three real datasets spanning a 36 $\times$ density range (Table 2): MovieLens-1M [38], Gowalla check-ins [39], and the Video Games category of the Amazon Reviews corpus [40, 41]. All three carry genuine interaction timestamps, which is a requirement rather than a convenience: three of the five players (*rec*, *seq*, and the decay term of *pop*) are uninterpretable without them.

Timestamp-free processed splits are excluded because file order cannot define *pop*, *rec*, or *seq*; only original timestamped sources enter the study.

Gowalla and Amazon are subsampled to fit a memory budget, with per-corpus scale *derived* from that budget rather than chosen: five dense score matrices at the full Gowalla size would require roughly 129 GB.

Content for ct and rec Gowalla and Amazon do not provide the same item text as MovieLens, so content is derived from available metadata rather than treated as absent. Gowalla venues use nested geographic cells at roughly 100, 10, and 1 km (each emitted on two half-offset grids, so that venues either side of a cell boundary still share a token); Amazon items from the category path and store fields of the product-metadata dump. A structural audit now runs before any coalition is scored and refuses to proceed when a player cannot reorder any candidate list (Section 3.5).

Protocol Leave-last-out temporal split where timestamps exist; ties broken by (timestamp, record index). The split is chronological *within each user* and is not globally time-blocked, a distinction worth making explicit. Because every source is fitted on the pooled training fold, and because *pop* decays relative to the latest *global* training timestamp, training events belonging to other users can postdate a given user's held-out event. We measure the exposure, and it is not small. Every test event on all three corpora precedes the global training $t_{\max}$, and the mean fraction of pooled training events that postdate a user's test event is 44.2% on MovieLens-1M, 27.1% on Amazon-VG and 19.4% on Gowalla. The medians are lower on the sparser corpora (43.6%, 18.1%, 5.7%), so the exposure is concentrated in early users rather than spread evenly, but the ninetieth percentiles reach 87.9%, 68.4% and 61.4%. This is the standard leave-last-out protocol in the recommender literature and we keep it for comparability, but global-timeline leakage under this protocol is documented and can change model orderings [42]. Our numbers therefore describe an offline per-user chronological evaluation rather than a globally causal deployment simulation, and the temporally sensitive sources (*pop*, *rec*, *seq*) are the ones whose values that distinction most affects.

A globally blocked replication, and what it costs Splitting MovieLens-1M at the 0.90 and 0.95 global timestamp quantiles gives folds in which *no* training event postdates an evaluated event. Only 526 of 6 038 users survive (8.7%), candidate recall falls to 0.464, below the 0.60 gate this paper declares in Section 3.5, and $v(\mathcal{G})$ falls from 0.0522 to 0.0095.

The allocation ordering agrees with the main ordering at $\tau = 0.60$ and LOO_{rank} at $\tau = -0.40$; the material sign disagreement moves from *cf* to *pop* and *seq*. We also reran the end-to-end single-source removals: against the observed removal ordering, ranking-stage LOO attains $\tau = 0.80$ and Shapley $\tau = 0.60$, and LOO identifies the source with the smallest observed removal loss. Thus the allocation-removal conclusion survives globally causal chronology, but the named source attributions do not. Because the blocked run both changes the user population and fails the recall gate, it supports only a relative sensitivity statement, not a cleaner estimate of the same target population.

Ten training seeds (42–51) are used for every result that carries a run-to-run interval: the attributions, pairwise interactions, and retirement analysis. These runs share the same

Table 2

Corpus and candidate-set preconditions. Users/eval. reports retained and evaluated users; monotonicity entries are material/raw counts.

Corpus	Users / eval.	Items	Density	$N_{\max}$	Recall (test)	Val. recall	Monot. (mat./raw)	Prop. 2
MovieLens-1M	6 038 / 6 035	3 533	2.697%	600	0.748	0.799	7 / 26	no
Amazon-VG	7 120 / 7 120	3 516	0.469%	600	0.589 [†]	0.648	4 / 14	no
Gowalla	8 865 / 8 865	82 134	0.075%	11 623	0.450 ^{†‡}	0.462	6 / 13	no

[†]Below the 0.60 gate; reported under the ceiling restriction described in this section (relative contrasts only). [‡]Recall on this corpus is bounded above by $\rho = 0.501$, the fraction of users whose test venue is not already in their training history (Table 4); 0.450 is 0.898 of that attainable ceiling and the gate is unreachable at any $N_{\max}$. The corresponding ceilings are 1.000 for MovieLens-1M and 0.972 for Amazon-VG.

users, interactions, temporal split, and corpus subsample. Their intervals therefore quantify sensitivity to stochastic fitting only; they are *not* confidence intervals for a population of users, datasets, or deployments. Diagnostics that remain illustrative, namely the estimand comparison and the per-corpus robustness sweeps, use seed 42 and are labelled as such. Wilcoxon signed-rank [43], permutation tests with 10 000 shuffles, Holm–Bonferroni correction [44], and Cohen’s d_z are retained as conditional descriptions under the fixed datasets rather than population-level inference.

Reporting protocol All ranking comparisons evaluate *every* method over the full catalogue; our method scores items outside C_u as $-\infty$ so its recall ceiling is a visible cost rather than a hidden denominator advantage. Restricting the baseline to C_u would be disclosable but is simply a weaker experiment. Note the two-protocol split: $v(S)$ remains fixed-candidate by definition of the game; only reporting is full-catalogue. Fair recommendation comparisons also require a common protocol and matched tuning budgets [45]. The under-tuned neural models in Appendix A are therefore descriptive reference points only and are not used for superiority claims.

3.5. Preconditions

Table 2 carries the two gates, and on this corpus set both bite. Density is computed on the data as used. Users with an unretrieved validation item remain in the head fit and contribute a zero target; the sensitivity analysis of Section 3.2 tests that choice. Recall is computed over the users evaluated by the game: 6 035 of the 6 038 MovieLens users have a nonempty candidate set, whereas all retained Amazon-VG and Gowalla users do. The three empty MovieLens cases are excluded according to Eq. (4) and counted in the coverage diagnostic rather than inferred from the segmentation.

Monotonicity On all three corpora the fitted game is **non-monotone**, so Property 2 does not apply as stated and a negative φ_g must be read as substantive rather than as numerical error. We regard this as a genuine finding about coalition-conditional refitting rather than an inconvenience: adding a source to a coalition can make the refitted ridge weights worse for some users.

Table 2 reports two counts, and the distinction is not pedantry. The raw count is every audited pair with $v(S \cup \{g\}) < v(S)$; the material count keeps only those with $|v(S \cup \{g\}) - v(S)| > 10^{-3}$. Most violations are far below the scale of v itself (the median is 4.5×10^{-4} against $v(\mathcal{G}) \approx 0.05$), so the raw count is sensitive to floating-point ordering and is not a reproducible statistic. Identical computations on identical data reported 17, 23 and 26 raw MovieLens violations on three separate runs differing only in platform and candidate rule, while every one of them reported the same material count of 7. Amazon-VG behaves the same way, moving from 12 raw to 14 with its material count fixed at 4. That is precisely the intended division of labour between the two columns, and it is the reason the material count is the one the argument uses. The Property 2 argument is what we quote as the headline. Thus 10^{-3} is a declared reporting threshold chosen to exceed the observed cross-platform instability, not a statistical significance cutoff or a weakened theorem hypothesis.

To be precise about what each count licenses, since these are three different things. *Exact* monotonicity is the theorem’s hypothesis, and it fails on all three corpora at any tolerance: Property 2 does not apply, full stop, and no threshold recovers it. The *material* count answers a different question: whether the violations are large enough to matter for interpretation, which bears on how seriously to read a negative φ_g . The *raw* count answers neither reliably, because it is not reproducible. We report material and raw, use exact for the theorem, and do not use the threshold to weaken a mathematical hypothesis. It is the reason the monotonicity hypothesis was made explicit in Section 3.3 rather than assumed.

Candidate recall, and a limit we can locate Under the frozen primary configuration, MovieLens clears the 0.60 gate at 0.748, whereas Amazon-VG and Gowalla do not. Amazon-VG nevertheless clears the gate in the candidate-cap sensitivity without changing the qualitative attribution result; Gowalla fails for a different reason.

The value 0.60 is an engineering applicability threshold, not a statistical cutoff or theorem constant. It was fixed before inspecting the primary attribution outcomes and requires at least three fifths of evaluated targets to be reachable, limiting zero-by-construction payoffs to at most 40% of users. Its

sensitivity is transparent from the evaluated configurations: at a 0.50 gate the frozen MovieLens and Amazon-VG runs pass; at 0.60 and 0.70 the frozen MovieLens run and the Amazon-VG $N_{\max} = 1200$ sensitivity pass; at 0.80 none of the reported capped configurations passes. Thus the numerical gate changes which configurations support absolute interpretation, while the source signs and leading qualitative conclusions remain stable across the candidate-cap points actually evaluated.

For Amazon-VG, recall is 0.588 at $N_{\max} = 600$. That value is 17.1% of its catalogue, the same fraction MovieLens receives; a log fit to the measured recall curve puts the 0.60 crossing at $N_{\max} \approx 634$. We did not move the frozen $N_{\max}$, and the reason is methodological rather than technical: it is frozen in advance, and shifting it by six percent of a threshold *after* observing that threshold is selection on the very quantity the parameter gates.

Changing the headline configuration and adding a declared robustness point are different acts, though, and the candidate-cap sweep already contains the second. At $N_{\max} = 1200$ Amazon-VG reaches recall 0.750, comfortably clear of the gate and almost exactly MovieLens-1M's 0.748, and the attribution barely moves:

$$\begin{aligned}\phi_{600} &= (0.01926, 0.00187, 0.00157, -0.00047, 0.01909), \\ \phi_{1200} &= (0.01920, 0.00168, 0.00319, -0.00042, 0.01804).\end{aligned}$$

Here *cf* still first, *seq* still second, *rec* still the only negative source, and Kendall $\tau = 0.80$ against the frozen-configuration run, the single transposition being *pop* and *ct*, which are separated by 0.0003. So the qualitative reading of Amazon-VG does not depend on being below the gate: it holds at a cap where the corpus clears it. We keep $N_{\max} = 600$ as the reported configuration because it was declared first, and report the gate-clearing point as the robustness evidence it is. Amazon-VG's role is unchanged in the main tables, but a reader who requires a gate-clearing corpus now has two.

For Gowalla the situation is qualitatively different, and the largest single term is one we originally missed. Half of this corpus's evaluated users (49.90%, Table 4) have a test venue that already appears in their own training history, and the training-history mask makes such an item ineligible in every candidate set. Candidate recall is therefore bounded above by $\rho = 0.501$ on this corpus, and the 0.60 gate is unreachable at *any* $N_{\max}$, including the full catalogue. The measured 0.450 is 0.898 of its attainable ceiling, which is a better score than Amazon-VG's 0.605 conditional value and not far below MovieLens-1M's 0.748, where the ceiling is 1.000. The gate as declared compares a quantity against a threshold that the corpus's own repeat structure forbids it from reaching; we retain the gate and Gowalla's failure of it as declared, but the reason is mask-versus-repeats, not only sparsity. The sweep below should be read against the 0.501 ceiling rather than against 1.0.

We did raise $N_{\max}$, from 2500 to 11623. The first value was inherited from a differently sized split and amounted

to only 3.04% of this corpus's catalogue, against 17% elsewhere, so the raise is justified by consistency rather than by outcome. We note for exactness that 11623 was derived from a 68443-item extract of this corpus and is 14.2% of the 82134-item version finally used, close to but not identical with the 17% the rule prescribes. We did not re-derive it after the corpus grew, precisely because adjusting $N_{\max}$ once recall is known is the selection we are trying to avoid. Recall improved to 0.450. The robustness sweep then measured the curve directly: $N_{\max} = 5811 \rightarrow 0.418$, $11623 \rightarrow 0.450$, $23246 \rightarrow 0.471$. Doubling the pool buys +0.021.

Recall is bounded above by ρ , the fraction of users whose test item is not already in their training history, and by nothing tighter: every *unseen* item receives a finite score from every source, so setting $N_{\max}$ to the catalogue size makes C_u the whole eligible catalogue and recall approaches ρ by construction. On MovieLens $\rho = 1.000$ and $N_{\max} = 3533$ yields recall 0.9995, as that reading predicts. On Gowalla $\rho = 0.501$, so the sweep is asymptoting toward 0.501 rather than toward one, and the 0.60 gate is unreachable at any pool size. This conclusion does not rest on extrapolating the logarithmic recall fit beyond its sampled range, which would not license it; it follows directly from the measured masking rate.

The remaining measurements establish a cost on top of that ceiling. Even reaching 0.501 on Gowalla would require a candidate set large enough that the fixed-candidate design stops being a retrieval-plus-ranking decomposition and becomes full-catalogue ranking, at which point the game no longer models the two-stage architecture it was built to explain, and the 32 coalition refits become correspondingly expensive. The practical statement combines a hard ceiling with an affordability limit below it: on a corpus this repetitive the gate is unreachable outright, and on a corpus this sparse the candidate pool needed for a usable recall ceiling is a large fraction of the catalogue, and a practitioner should check that ratio before adopting a fixed-candidate game. Quantifying where that tradeoff becomes unacceptable, and whether recall rises fast enough in the unsampled region between 23246 and 82134 to change the assessment, requires a full-catalogue sweep we have not run.

What the ceiling restriction means Both affected corpora are retained for *relative* comparisons (the LOO-versus-Shapley contrast of Section 4.3, which is a difference between two attributions computed on the same candidate sets) and excluded from absolute NDCG@10 comparison against the neural baselines, where a suppressed recall ceiling would flatter or penalise methods unequally. This follows the fall-back ladder fixed before these runs. A failed-gate corpus is admitted only if it appears in a declared exemption list, and every associated table carries the restriction explicitly.

For Gowalla this restriction is stronger. Repeated venues are valid next-location outcomes, yet the present protocol masks them because they occur in training. Until a repeat-aware sensitivity is run, Gowalla is treated only as a relative stress test of the attribution estimands under repeat masking;

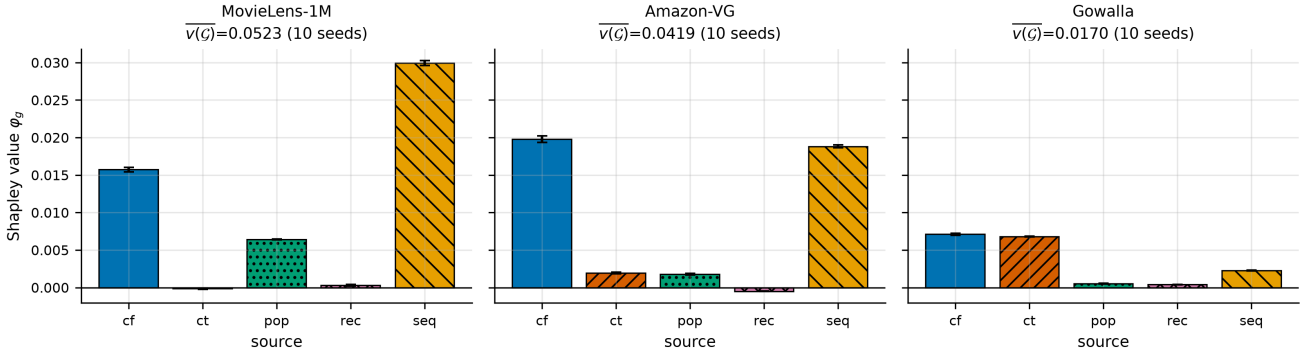


Figure 2: Per-source Shapley values over seeds 42–51. Whiskers are $\bar{x} \pm t_9 \text{SE}$ run-to-run intervals.

it is not primary evidence about next-location recommendation.

Per-source attributions with seed-based intervals appear in Figure 2. Efficiency holds to machine precision on every corpus, with worst-case $|\sum_g \phi_g - v(G)| = 6.9 \times 10^{-18}$.

Configuration Table 3 lists every value that affects a reported number. All are fixed before the runs and are not tuned per corpus or per coalition; where a value was amended after the first runs, the date and reason were recorded. We report the values directly because a frozen configuration is not a substitute for a stated one.

Preprocessing and subsampling MovieLens-1M is binarised at rating ≥ 4 , which is why the as-used interaction count is 575 276 rather than one million. Gowalla and Amazon-VG are filtered to a 10-core and binarised (Amazon at rating ≥ 4 ; Gowalla check-ins are implicit). Both are then subsampled by drawing users uniformly at random without replacement under seed 42, retaining all interactions of the drawn users; the number of users is *derived* from a declared memory budget rather than chosen, since five dense score matrices at 8865×82134 already occupy 14.6 GB and the full corpus would need 129 GB. Item vocabularies are re-indexed after subsampling, so the catalogue sizes in Table 2 are post-subsampling.

4. Results and discussion

4.1. Implementation checks against known axiomatic constraints

On observational data the correct allocation is unknown. This section therefore does not validate the real-system magnitudes; it checks whether the implementation satisfies constraints known in advance. Duplicate injection fixes an exchangeability requirement, while the analytic games check exact magnitudes for synthetic characteristic functions.

Design We inject a near-duplicate of an existing source, $g_{\text{dup}} = g + \eta \sigma_g \epsilon$ with ϵ standard normal, and re-run the game

with six players. The candidate set is held *fixed* at its pre-injection value, so the only thing that changes is the player set.

What is known in advance, and what is not At $\eta = 0$ the original and duplicated sources are exchangeable in the declared game, so any rule satisfying the symmetry axiom must assign them equal values. This gives a checkable constraint that does not depend on the data. Property 2 suggests a second outcome: leave-one-out should collapse toward zero for *both* members, since either alone is dispensable. **This one is not structurally guaranteed under our fitted game, and we do not treat it as such.** Duplicating a column in a ridge regression is not prediction-invariant: two identical columns share the penalty, so the effective regularisation on their common direction is halved and the fitted scores can move. Exchangeability of the pair is guaranteed; redundancy of the coalition *values* is an empirical question. We therefore measure it, reporting

$$\Delta_{\text{red}} = \max_{S \subseteq G \setminus \{g, g_{\text{dup}}\}} \left\{ \begin{aligned} &|v(S \cup \{g\}) - v(S \cup \{g, g_{\text{dup}}\})|, \\ &|v(S \cup \{g_{\text{dup}}\}) - v(S \cup \{g, g_{\text{dup}}\})| \end{aligned} \right\}. \quad (14)$$

alongside the symmetry error, and we run the diagnostic at both the frozen λ and at $\lambda = 0$. The latter matters because ordinary least squares *is* invariant to a repeated column, since with the minimum-norm solution a shared coefficient splits evenly and predictions are unchanged, so $\lambda = 0$ turns the redundancy prediction into a structural guarantee and isolates whether any deviation at the frozen λ is due to regularisation.

We are deliberately precise about the limits of this design. **Duplication does not conserve credit**, and it would be wrong to require that it should. The combined Shapley value of a cloned pair need not equal the original player's pre-duplication value: in the two-player monotone game $v(g) = 7$, $v(h) = 8$, $v(g, h) = 8$ one has $\phi_g = 3.5$, whereas replacing g by perfect substitutes g_1, g_2 gives $\phi_{g_1} = \phi_{g_2} = 7/3$ and a pair sum of $14/3 \approx 4.67$. Clone insertion redistributes credit among *all* players, a sensitivity of Shapley

Table 3

Experimental configuration.

Component	Parameter	Value	Note
<i>cf</i>	latent factors	64	ALS, implicit feedback binarised r_{ui}
	confidence scaling	$\alpha = 1, c_{ui} = 1 + \alpha r_{ui}$	
	regularisation	0.1	
	alternating iterations	15	
<i>ct</i>	TF-IDF vocabulary	5000	token pattern \s+
<i>pop</i>	decay half-life	180 d	exponential
<i>rec</i>	content clusters	20	argmax over a 20-component truncated SVD of a separate 2000-vocabulary TF-IDF item matrix (no k -means); <code>random_state = seed</code>
<i>seq</i>	TF-IDF vocabulary	2000	separate item matrix from <i>ct</i> 's 5000
	decay half-life	30 d	
	embedding dimension	64	PPMI-SVD
	PPMI shift	$k_s = 1$, so $\log k_s = 0$	unshifted: the configuration is plain PPMI-SVD
	co-occurrence window	5	
Game	$N_{\max}$	600 / 600 / 11 623	ML-1M / Amazon-VG / Gowalla
	ridge λ	$1.0^\dagger$	
	NDCG cutoff K	10	summed-loss parameter; effective averaged-loss value is $1/ U_{\text{fit}} $
	empty-coalition baseline	$\mathbb{E}[\text{NDCG}@10]$	
	ranking tie rule	score, then item index	Equation (7); deterministic, no seed exact total order; no tolerance relation
	initial per-source depth	$\lceil N_{\max}/n \rceil$	
	candidate growth iters	10	$N_g \leftarrow N_g + \lceil (N_{\max} - C_u)/n \rceil$ per pass
	recall gate	0.60	
	materiality threshold	10^{-3}	see Section 3.5
			reporting only, not a theorem
Fusion	segments	4	activity $\times$ recency, median splits
	softmax τ	$\{0.001, 0.01, 0.1\}^\dagger$	validation-selected
	shrinkage γ_{shrink}	$\{0, 0.5, 1\}^\dagger$	validation-selected
LightGCN	factors / layers	64 / 3	contextual reference only
SASRec	epochs / lr / reg	40 / 0.01 / 10^{-4}	contextual reference only
	dim / max length	64 / 50	
	blocks / epochs / lr	2 / 25 / 0.05	
Protocol	seeds	42–51 (run-to-run summaries)	seed 42 alone for labelled single-run diagnostics passed to ALS and every truncated SVD test = last, validation = second-last
	source random state	seed	
	split	leave-last-out	
	tie-break	(timestamp, record index)	

[†]Selected on a MovieLens pilot and then frozen. All other values are fixed defaults; $N_{\max}$ alone varies by corpus.**Table 4**

Repeat-event audit. Percentages are of users holding both folds. “Test item in train” users are masked out of every candidate set, so their per-user payoff is zero for all 32 coalitions.

Corpus	Users	Val. = test item	Test item in train	Repeat events
MovieLens-1M	6 035	0 (0.00%)	0 (0.00%)	0.00%
Amazon-VG	7 120	430 (6.04%)	198 (2.78%)	5.27%
Gowalla	8 865	1 260 (14.21%)	4 424 (49.90%)	52.61%

allocations to replication that is known in the cooperative-game literature [29]. Our own runs show the same effect: on MovieLens-1M the pair sum is 0.02156 against a pre-injection $\phi_{cf} = 0.01520$ in the same injection run, a ratio of 1.42. We therefore treat the pair sum as a *diagnostic*, not a requirement, and the experiment as a symmetry and implementation check rather than as recovery of absolute credit.

Table 5 reports the outcome. The result is unambiguous and reproducible: Shapley satisfies symmetry exactly in every condition. LOO returns **exactly zero** for both members of every pair, not approximately but exactly, because $v(\mathcal{G}) =$

$v(\mathcal{G} \setminus \{g\})$ holds numerically when a perfect substitute remains in the coalition. Shapley splits the shared capability evenly, giving each clone 0.01078 against 0.01520 for *cf* before injection, a pair sum of 0.02156: the split is exact, and the pair is worth more together than the single source was, which is the non-conservation discussed above.

The practical reading needs care. It would overstate the result to say an engineer would retire both duplicates and lose the capability. A sequential procedure would re-evaluate the system after the first removal and would not make that error. The narrower point, which is the one the diagnostic establishes, is that grand-coalition LOO does not *distribute*

Table 5Duplicate-source diagnostics on MovieLens-1M (cf clone, $\eta = 0$).

Corpus	λ	Sym. err.	Δ_{red}	$\varphi(cf) = \varphi(cf_{\text{dup}})$	LOO(cf), LOO(cf_{dup})
MovieLens-1M	frozen	0.0	0.0	0.010781	0.000000, 0.000000
MovieLens-1M	0	0.0	0.0	0.010781	0.000000, 0.000000

Table 6

Exact Shapley recovery on six analytic games.

Game	Characteristic function $v(S)$	Expected φ	Max error
Additive	$\sum_{g \in S} c_g, \quad c = (3, 2, 1)$	(3, 2, 1)	0
Substitutes	$4 \cdot \mathbb{1}[\{s_1, s_2\} \cap S \neq \emptyset] + \mathbb{1}[x \in S]$	(2, 2, 1)	0
Complements	$6 \cdot \mathbb{1}[\{c_1, c_2\} \subseteq S] + 2 \cdot \mathbb{1}[x \in S]$	(3, 3, 2)	0
Null player	$\sum_{g \in S \cap \{a, b\}} c_g, \quad c = (3, 2)$	(3, 2, 0)	0
Harmful	$5 \cdot \mathbb{1}[a \in S] - 2 \cdot \mathbb{1}[bad \in S]$	(5, -2)	0
Unanimity	$9 \cdot \mathbb{1}[S = \mathcal{G}]$	(3, 3, 3)	0

shared credit: it can report zero for each of two perfect substitutes while the capability they share is valuable, so it cannot be used as a simultaneous allocation rule. Shapley symmetry assigns exchangeable players equal credit and does not have this failure. Here that difference is measured rather than argued.

4.2. Verifying the aggregation

Analytic games with known Shapley vectors Duplicate injection tests symmetry, but symmetry alone does not validate *magnitudes*: a method that halved every value would still pass it. We therefore also evaluate six analytic games whose Shapley vector is known in closed form, chosen to span the structures our real games might exhibit (Table 6). These are unit tests of the aggregation, not evidence about recommender data; they establish that the enumeration in Equation (9) is implemented correctly, which is a precondition for interpreting anything computed with it.

We additionally verify *additivity*, the remaining uniqueness axiom that no other diagnostic here exercises: efficiency is a sum constraint, symmetry is covered by duplicate injection, and the null player has its own game. Summing an additive and a unanimity game, structurally different so the check is not vacuous, gives $\varphi(v_1 + v_2) = \varphi(v_1) + \varphi(v_2)$ to 4.4×10^{-16} .

The null-player and harmful games matter most for this paper. Our fitted games are non-monotone and produce negative φ_g (Section 4.3), so it is necessary to establish that a negative value is a faithful computation rather than an artefact of the implementation; the harmful game confirms that -2 is recovered exactly. The null-player game confirms that a source contributing nothing receives exactly zero, which is the property the structural audit of Section 3.4 relies on.

Alternative values and pairwise interactions

Other semivalues The Shapley value is one member of a family, and a comparator is only informative if it is genuinely

a different member. The size-uniform weighting $p_k = 1/n$ is not: substituting it into Equation (1) returns Equation (9) exactly, because $|S|!(n - |S| - 1)!/n! = 1/(n \binom{n-1}{|S|})$. It is Shapley written by coalition size, it satisfies efficiency, and comparing it against Shapley is vacuous. We note this because the identity is easy to miss.

The distinct comparators we report are Banzhaf, $p_k = \binom{n-1}{k} 2^{-(n-1)}$, and the binomial semivalues $p_k = \binom{n-1}{k} q^k (1-q)^{n-1-k}$ at $q = 0.25$ and $q = 0.75$, in which each other source joins the predecessor coalition independently with probability q . Small q favours sources that are valuable in isolation; large q favours sources that still add value once the others are present, so the pair brackets the redundancy question the paper is about. On MovieLens (seed 42) all four values induce the *same* source ordering, with *seq* first (Kendall $\tau = 1.00$ against Shapley in each case), while their totals differ widely, 0.05190 for Shapley against 0.05167, 0.06992 and 0.03378 for Banzhaf, $q = 0.25$ and $q = 0.75$. These four totals are a within-platform set computed on x86-64, so the Shapley entry is the 0.05190 of Section 3.1 rather than the 0.05136 seed-42 main-game value underlying the refitted-head row of Table 11; only their ratios are meaningful, and the point of the comparison is that the *ordering* is identical while the totals are not. Only Shapley sums to $v(\mathcal{G})$; for the others only the induced ordering is comparable. We computed this on MovieLens only, so the claim is that the MovieLens ordering is robust to the choice among these values, and we do not extend it to the other two corpora.

Pairwise interactions Main-effect values cannot by themselves distinguish a source of low value from a source whose value is absorbed by a redundant partner. We therefore compute a pairwise interaction index: the classical *Shapley interaction index* of Grabisch and Roubens [24]. This index differs from the order-2 Shapley–Taylor index [25]: the two are not interchangeable, since at $n = 5$ their coalition weights differ by up to a factor of 2.5. We use Grabisch–Roubens throughout, and note that Grabisch–Roubens does

Table 7

Pairwise Shapley interactions on MovieLens-1M. Intervals describe variation over ten stochastic fits.

Pair	I_{ab} (10-seed mean)	run interval	Neg. seeds
<i>cf-seq</i>	-0.0542	[-0.0554, -0.0529]	10/10
<i>cf-pop</i>	-0.0254	[-0.0256, -0.0252]	10/10
<i>ct-pop</i>	-0.0027	[-0.0029, -0.0025]	10/10
<i>pop-rec</i>	-0.0017	[-0.0019, -0.0016]	10/10
<i>cf-rec</i>	-0.0011	[-0.0014, -0.0009]	10/10
<i>rec-seq</i>	-0.0011	[-0.0014, -0.0009]	10/10
<i>ct-rec</i>	+0.0004	[+0.0003, +0.0006]	0/10
<i>ct-seq</i>	+0.0012	[+0.0009, +0.0014]	0/10
<i>cf-ct</i>	+0.0020	[+0.0016, +0.0023]	0/10
<i>pop-seq</i>	+0.0059	[+0.0056, +0.0061]	0/10

not satisfy Shapley–Taylor’s interaction-efficiency axiom: the pairwise terms do not sum with the singletons to $v(\mathcal{G})$. That is acceptable here because we use them only to *rank* redundancy, which is all the argument requires. Each pair $\{a, b\}$ receives

$$I_{ab} = \sum_{S \subseteq \mathcal{G} \setminus \{a, b\}} \frac{|S|!(n - |S| - 2)!}{(n - 1)!} \times [v(S \cup \{a, b\}) - v(S \cup \{a\}) - v(S \cup \{b\}) + v(S)], \quad (15)$$

the averaged second-order discrete derivative. A negative I_{ab} means the pair jointly contributes less than the sum of their separate contributions. We read that as negative, or substitutive, interaction *under the declared game*, which is a game-theoretic statement and not a demonstration of causal redundancy; a positive value indicates complementarity in the same sense.

Table 7 localises the negative interactions associated with the MovieLens sign flip. The two most negative pairs are *cf-seq* (−0.054) and *cf-pop* (−0.025), both stable in sign on all ten seeds: collaborative filtering has the strongest substitutive interaction with the sequential and popularity sources, which is why removing it from the grand coalition costs so little and why leave-one-out scores it below zero. Only one of these, *cf-pop*, was declared in advance (Section 3.4); *cf-seq* was not, and the other declared overlap *rec-ct* comes out slightly *positive* at +0.0004. We therefore claim partial rather than complete recovery of the pre-declared structure. Every pair in Table 7 keeps its sign on all ten seeds, so the interaction ordering does not rest on a single run.

The redundancy mechanism replicates on Gowalla The interaction index is cheap once the 32 coalition values exist, so we computed it on the other two corpora as well; those are seed-42 values under the legacy candidate rule and we label them as such. The test is whether the source carrying each material sign disagreement is also the one with the strongest substitutive interaction. It is. On Gowalla the most negative pair is *ct-pop* at −0.0083, and *pop* is the source whose ranking-stage leave-one-out turns negative, exactly as *cf* is on MovieLens-1M, where the two most negative pairs both involve *cf*. On Amazon-VG, which has

no material disagreement, the most negative pair is *cf-seq* at −0.0445 while *cf* carries the largest positive gap rather than a sign flip, so a strong substitutive interaction is not by itself sufficient to produce one. The mechanism is therefore consistent across the two corpora that exhibit the phenomenon, and the negative case shows the interaction index alone does not predict it.

4.3. Attribution on real systems

Table 8 gives the per-source comparison and Figure 3 plots it. Material sign disagreement occurs on MovieLens and Gowalla, while Amazon-VG provides a negative case.

What is consistent, and what is not Two of the three corpora contain a *material* sign disagreement, one where the gap exceeds 10^{-3} , the same threshold we apply to monotonicity. On MovieLens it is *cf*: Shapley scores it +0.01573, the second most valuable source in the system, where LOO_{rank} scores it −0.00457, meaning its removal *improves* the fixed-candidate ranking game. The gap is 0.02030 [+0.02007, +0.02053], or 38.8% of the ten-seed $v(\mathcal{G}) = 0.05225$, and it is positive on 10/10 seeds. On Gowalla it is *pop*, at 12.2% of $v(\mathcal{G}) = 0.01705$ and likewise 10/10: LOO_{rank} scores the popularity signal negative where Shapley scores it positive. All figures in this paragraph are ten-seed means, so the gap and the grand value are at the same aggregation level; the brackets describe run-to-run variation over stochastic fits on the fixed datasets.

Amazon-VG has no material disagreement. Its only sign flip, on *rec*, is 0.00063, a practically negligible magnitude even though the Shapley run-to-run interval excludes zero. We state this plainly because it bounds the claim: it would be wrong to say that *every* system contains a sign disagreement. The accurate statement is weaker: material sign disagreement between the two rules occurs on two of the three systems studied here and is absent on the third. Three corpora cannot estimate how common it is in the wider population of hybrid recommenders; what they establish is that it is possible and can be large. Nothing in either rule announces in advance whether a given system will exhibit it.

Within-run regularisation sensitivity At seed 42, we swept the summed-loss parameter λ over eight orders of magnitude, 10^{-5} to 10^3 , recomputing the full game at each value. The material flips are invariant: *cf* flips on MovieLens and *pop* flips on Gowalla at *every* λ tested, and they are the only material flips on those corpora throughout. Amazon-VG has none until $\lambda = 10^3$, where the head is so heavily penalised that *ct* acquires one; we regard that endpoint as degenerate rather than informative. This establishes within-run sensitivity to the chosen penalty, not cross-seed or cross-corpus invariance. Figure 4 shows the representative points 0.1, 1, and 10; the conclusions above refer to the complete evaluated grid. Because Eq. (8) uses a summed loss, the numerical values are not portable across corpus sizes.

Table 8

Ranking-stage leave-one-out and Shapley attribution over ten seeds. Brackets are run-to-run intervals, $\bar{x} \pm t_{0.975,9} s_x / \sqrt{10}$. The quantity this paper contrasts is the paired gap $\Delta_g = \varphi_g - \text{LOO}_{\text{rank}}(g)$, so it carries its own interval, computed per seed and then aggregated rather than differenced from the two marginal means. The final column counts the fits on which the gap is positive. Fourteen of the 15 cells are unanimous; the exception is *rec* on MovieLens-1M at 9/10, whose gap is 0.4% of $v(G)$ and which the paper does not interpret. Both material disagreements have gap intervals excluding zero, at 38.8% (*cf*, MovieLens-1M) and 12.2% (*pop*, Gowalla) of the respective $v(G)$.

Corpus	Source	LOO _{rank}	Shapley [run interval]	Gap $\varphi_g - \text{LOO}_{\text{rank}}$ [run interval]	Positive-gap seeds	
MovieLens-1M	<i>cf</i> [†]	−0.00457	+0.01573 [+0.01542, +0.01604]	+0.02030 [+0.02007, +0.02053]	10/10	
	<i>ct</i> [‡]	+0.00052	−0.00016 [−0.00024, −0.00008]	−0.00068 [−0.00088, −0.00047]	0/10	
	<i>pop</i>	+0.00070	+0.00643 [+0.00638, +0.00649]	+0.00573 [+0.00551, +0.00596]	10/10	
	<i>rec</i>	+0.00010	+0.00032 [+0.00022, +0.00042]	+0.00022 [+0.00003, +0.00040]	9/10	
	<i>seq</i>	+0.01745	+0.02993 [+0.02963, +0.03024]	+0.01249 [+0.01209, +0.01288]	10/10	
Amazon-VG	<i>cf</i>	+0.00890	+0.01980 [+0.01937, +0.02022]	+0.01089 [+0.01062, +0.01117]	10/10	†Material
	<i>ct</i>	+0.00011	+0.00195 [+0.00183, +0.00207]	+0.00183 [+0.00165, +0.00201]	10/10	
	<i>pop</i>	+0.00394	+0.00178 [+0.00166, +0.00190]	−0.00216 [−0.00244, −0.00188]	0/10	
	<i>rec</i> [‡]	+0.00014	−0.00049 [−0.00053, −0.00046]	−0.00063 [−0.00069, −0.00057]	0/10	
	<i>seq</i>	+0.00761	+0.01883 [+0.01865, +0.01900]	+0.01122 [+0.01107, +0.01138]	10/10	
Gowalla	<i>cf</i>	+0.00743	+0.00711 [+0.00698, +0.00725]	−0.00032 [−0.00039, −0.00025]	0/10	
	<i>ct</i>	+0.00547	+0.00678 [+0.00672, +0.00683]	+0.00131 [+0.00121, +0.00141]	10/10	
	<i>pop</i> [†]	−0.00156	+0.00051 [+0.00046, +0.00056]	+0.00207 [+0.00197, +0.00217]	10/10	
	<i>rec</i>	+0.00001	+0.00040 [+0.00038, +0.00042]	+0.00039 [+0.00037, +0.00041]	10/10	
	<i>seq</i>	+0.00093	+0.00225 [+0.00219, +0.00230]	+0.00132 [+0.00125, +0.00138]	10/10	

sign flip; [‡]sign flip below the 10^{-3} materiality threshold.

Data-level resampling beyond training seeds Two sample-level robustness checks are available for the Gowalla result. The corpus was run at three subsample sizes, 4 652, 5 910 and 8 865 users, drawn to different memory budgets. The material *pop* disagreement is present at all three, and *pop* is the only material flip in each case: at the two larger sizes it is 9.9% and 12.2% of $v(G)$. A $1.9\times$ change in corpus size therefore does not create or destroy it. We report the largest run throughout.

The magnitudes are *not* stable, however, and neither is the top source. We report this against our own convenience. A full ten-seed repetition at a smaller memory budget draws 4 652 of 8 865 users and 59 597 of 82 134 items, which also raises the frozen $N_{\text{max}} = 11\,623$ from 14.2% to 19.5% of the catalogue. Under that resampling every Gowalla sign is preserved and the material *pop* flip survives, but individual values move by more than a factor of two, Kendall τ against the full run falls to 0.80, and the top-ranked source exchanges between *cf* and *ct*, the two sources the full run separates by only 0.00033. We therefore do not claim a stable top source on Gowalla, and the *cf* > *ct* ordering reported above should be read as a near-tie rather than a finding. The same repetition on MovieLens-1M and Amazon-VG, whose user caps are not budget-bound and whose corpora are therefore identical across the two runs, preserves the ordering exactly ($\tau = 1.00$, same top source). The conclusion is narrow: on the sparsest corpus, which is also the one that fails the recall gate by the widest margin, only the signs and the material disagreement survive resampling.

The corollary is that a single-corpus reading would have been actively misleading. On MovieLens alone one would

conclude that collaborative filtering is the source ablation systematically undervalues. Gowalla’s material flip falls on *pop* instead, and Amazon-VG has none. The results are consistent with the *mechanism* that disagreement tracks redundancy, though the interaction analysis supports that reading directly only on MovieLens. What certainly does not generalise is the identity of the affected source, or the guarantee that one exists at all.

Mechanism The account is *qualitatively* the one Property 2 describes, though the property does not formally apply: all three fitted games are non-monotone (Table 2), so the positivity conclusion is unavailable and what follows is interpretation rather than theorem. A source whose signal is covered by others costs little when removed from the *grand* coalition, because the others absorb its role, and ablation observes only that single contrast. SignalShap averages the source’s contribution over all 16 coalitions in which it can appear, including those where nothing covers for it, and so recovers value that the overlap conceals. On MovieLens the collaborative signal overlaps with the sequential and popularity signals; on Gowalla it is the popularity signal that is covered, by a geographic content signal that is essentially tied with *cf* at the top of the ranking (0.00678 against 0.00711, an ordering that resampling reverses).

Ranking stability The per-corpus orderings are *seq* > *cf* > *pop* > *rec* > *ct* (MovieLens), *cf* > *seq* > *ct* > *pop* > *rec* (Amazon-VG), and *cf* > *ct* > *seq* > *pop* > *rec* (Gowalla, where the leading pair is within 0.00033 and exchanges under resampling). Pairwise Kendall τ between

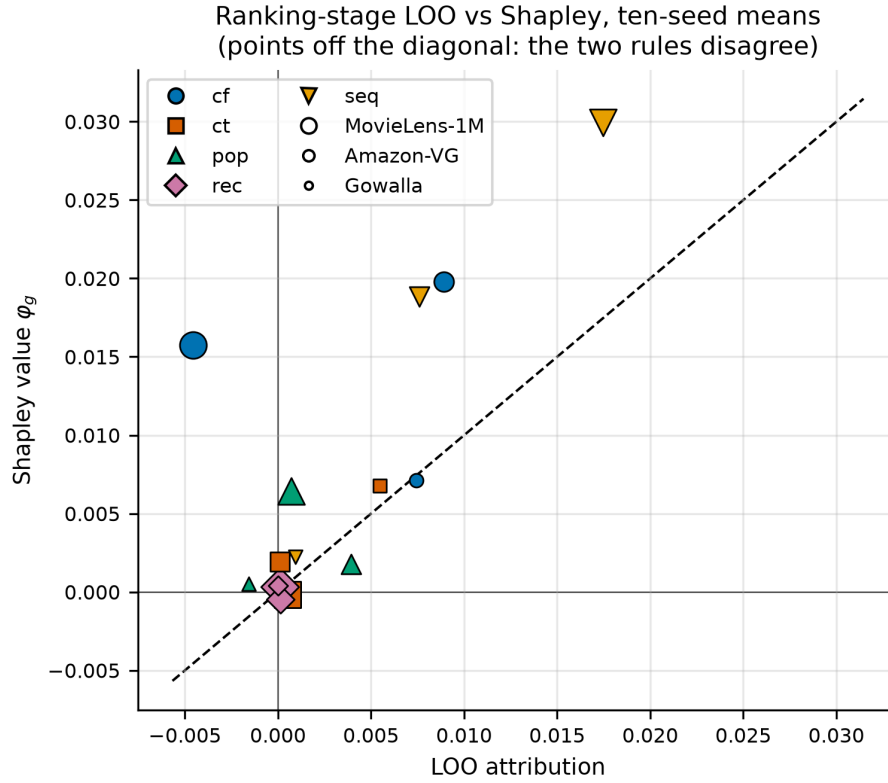


Figure 3: Ten-seed mean Shapley and ranking-stage leave-one-out attribution; the diagonal denotes equality.

corpora is 0.40 (MovieLens–Amazon), 0.20 (MovieLens–Gowalla) and 0.80 (Amazon–Gowalla), none significant at $n = 5$ sources. We therefore make no cross-corpus consistency claim; identical near-zero values from sources with no usable input would inflate this comparison and are not treated as evidence of agreement.

Score-vector association has a descriptive correlation of $r = 0.525$ with the attribution shift on MovieLens across the ten source pairs. This is not a test of Lemma 1: the lemma defines substitutability through coalition values, whereas the auxiliary statistic compares per-user score rankings. The ten pair observations also share sources and are not independent. We therefore report the coefficient only as an exploratory description, without a redundancy or significance claim.

Figure 5 shows the pairwise structure directly.

4.4. Explainability scope and diagnostics

SignalShap explains a fitted system at the level of architectural sources. The global values φ_g divide baseline-centred ranking utility, while $\varphi_g(u)$ provides the corresponding per-user decomposition. Faithfulness is therefore relative to the declared characteristic function: efficiency and the per-user decomposition are exact identities, and the analytic games verify the aggregation implementation. Pairwise interaction indices add information about substitutive or complementary behaviour under that same game, but they are not causal redundancy estimates.

The study does not evaluate whether end users understand or prefer these explanations, and it does not establish seed stability of individual-user attributions. Segment profiles in Appendix A summarize heterogeneity, but no segment contrast survives Holm correction. The supported explainability claim is consequently system-diagnostic source attribution, not human-facing explanation quality.

4.5. Ablation and sensitivity summary

The main design sensitivities are consolidated in Table 9; all are described in detail in their referenced sections.

4.6. Limitations

Analytic and data-independent, and what is not Two results here are mathematical and depend on no corpus: the counterexample refuting unconditional Property 2, and the $1/|G|$ constant in Lemma 1. The arguments against grand-coalition candidate pools and against per-coalition λ tuning are *not* of that kind. Neither is proved, so they should be read as construct-design choices rather than theorems. We do not show that a G -selected pool must raise $v(G)$ for an arbitrary relevance distribution, nor that per-coalition tuning must produce an optimism term increasing in $|S|$. They are construct-design choices: each removes a confound between the contribution of a source and a change in the retrieval pool or the model-selection procedure. We support the first with a measurement on MovieLens (Section 3.1) and the second with the λ sweep, which is evidence rather than proof.

Exact-enumeration ranking-stage source attribution

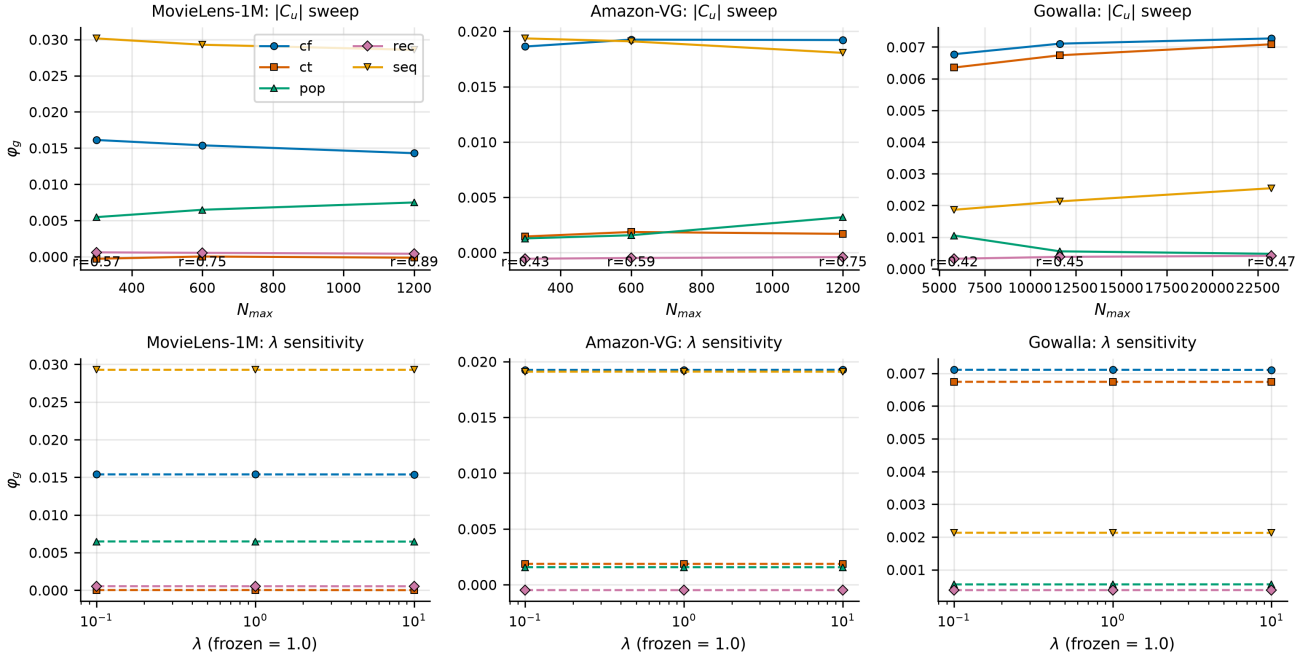


Figure 4: Seed-42 sensitivity to candidate cap (top) and ridge penalty (bottom); annotations report candidate recall.

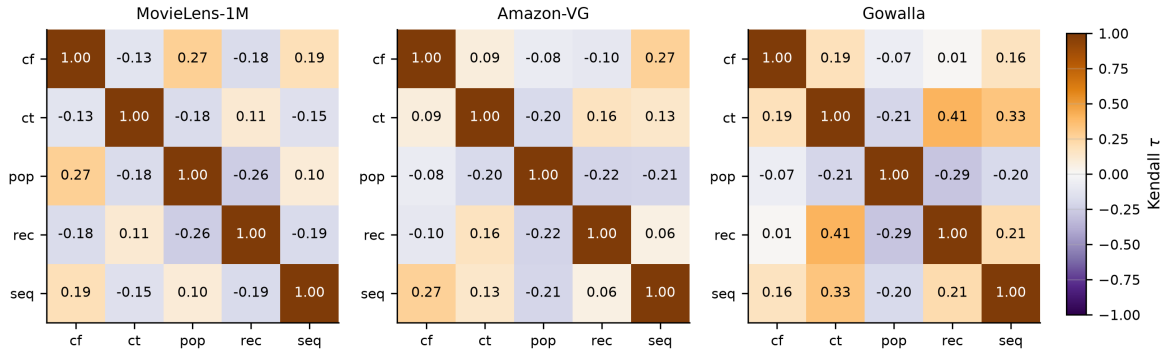


Figure 5: Pairwise source-score association across the three corpora (Kendall τ).

Table 9

Ablation and sensitivity summary.

Design contrast	Scope	Main outcome
Union vs grand-coalition pool	MovieLens, seed 42	Recall $0.748 \rightarrow 0.807$; ϕ_{ct} changes sign (Sec. 3.1).
Symmetric vs source-ordered truncation	MovieLens probe	Maximum attribution change 2.8×10^{-5} ; ordering unchanged (Sec. 3.1).
Retain vs drop validation misses	Three corpora, seed 42	All signs unchanged; maximum shift 2.1×10^{-4} (Sec. 3.2).
Frozen vs refreshed history	Three corpora, seed 42	$v(G)$ changes by 36%, 19%, and 3.7%; ordering changes on Amazon-VG (Sec. 3.2).
Per-user vs globally blocked time	MovieLens, seed 42	Named source attributions change; retirement agreement remains higher for LOO_{rank} (0.80 vs 0.60), but recall is 0.464 (Sec. 3.4).
Ridge-penalty sweep	Three corpora, seed 42	The two material sign disagreements persist over 10^{-5} – 10^3 (Sec. 4.3).
Gowalla user resampling	Alternative sample, ten seeds	Signs and the material disagreement persist; source-order agreement is $\tau = 0.80$, while magnitudes are unstable (Sec. 4.3).

Table 10

Provenance of reported result groups.

Result	Seeds	Candidate rule	Role
Attribution and retirement	42–51	source-symmetric	primary
Pairwise interactions	42–51	source-symmetric	explanatory
Estimand comparison (Table 11)	42	final main rows; legacy alternatives	sensitivity
Neutral candidate pools	42	all four pool rules	sensitivity
Preconditions and duplicate test	42–44	source-symmetric*	sensitivity
Sweeps, ablations, and segments	42–44	source-symmetric*	exploratory

*MovieLens-1M and Amazon-VG only. The Gowalla diagnostics remain under the legacy rule: regenerating them requires holding five dense 8865×82134 score matrices, which exhausted memory on the 48 GB machine used for these runs. Gowalla's primary attribution and retirement numbers are unaffected, since the ten-seed results use the source-symmetric rule on all three corpora.

Result provenance The ten-seed attribution and retirement results in this manuscript were regenerated under the source-symmetric candidate rule. Several labelled single-seed diagnostics predate that regeneration: candidate recall and monotonicity in Table 2, duplicate injection, segment profiles, the λ and candidate-cap sweeps, the grand-pool ablation, and the estimand comparison. On MovieLens, the old and symmetric rules agree to $\max_g |\Delta\phi_g| = 2.8 \times 10^{-5}$ with identical ordering and recall equal to three decimals. Accordingly, ten-seed results are primary; diagnostics that have not been regenerated under the symmetric rule are listed in Table 10 and are explicitly single-seed sensitivity analyses rather than confirmatory evidence.

Per-user rather than globally chronological The evaluation orders each user's own interactions in time but does not impose a global cutoff, so pooled training events can postdate another user's held-out event and are visible to sources fitted on that pool. We measure the exposure rather than assert it is negligible, and we ran a globally blocked replication on MovieLens-1M (Section 3.4). That replication is the sharpest limitation in the paper. Under a single global cutoff the per-source attributions do *not* reproduce: the allocation ordering agrees only at $\tau = 0.60$, LOO_{rank} at $\tau = -0.40$, and the material sign disagreement moves from *cf* to *pop* and *seq*. The qualitative claim survives, in that a material disagreement between allocation and ranking-stage removal is still present, but the particular sources it names do not. The blocked corpus also retains only 8.7% of users and fails the paper's own 0.60 recall gate, so it is a different and much smaller population rather than a cleaner measurement of the same one.

The paper's operational claim is specifically about retirement, though, and that claim is separable from the per-source allocation. We therefore ran the end-to-end retirement simulation under the blocked split as well. *It replicates*. Against the observed removal cost, LOO_{rank} attains $\tau = 0.80$ and the Shapley allocation $\tau = 0.60$, an advantage of $+0.20$ in the same direction as the main result; LOO_{rank} identifies the source with the smallest observed removal loss (*pop*), while the allocation names *rec* instead and is wrong. So the disagreement between allocation and removal, which is this paper's central claim, survives a globally causal split even on

a population where the individual attributions do not. The honest position is therefore split: our per-source numbers describe offline per-user chronological evaluation, which is what the recommender literature this work compares against also does, and we have not shown those transfer; the allocation-versus-removal contrast is the narrower finding that does.

Two-step-ahead evaluation from a frozen state The protocol of Section 3.2 freezes the source state at the training fold, so the validation interaction is neither added to the history nor masked before test scoring. We measured the cost rather than reasoning about it, and it is substantial on the dense corpora: a refreshed one-step-ahead protocol raises $v(\mathcal{G})$ by 36% on MovieLens-1M, 19% on Amazon-VG and 3.7% on Gowalla at seed 42 (across ten seeds the MovieLens figure is $+31.7\%$ [$+30.2\%$, $+33.1\%$]), most of it accruing to *seq*. The absolute coalition values reported here are therefore specific to the frozen protocol and understate what these sources achieve with a current history. The source ordering is unchanged on MovieLens and Gowalla but not on Amazon-VG, where the leading pair exchanges. It would be unsound to conclude from this that the frozen values are a lower bound on the refreshed ones: the measured direction agrees on all three corpora, but the argument ignores the simultaneous changes to the source scores, the candidate set and the baseline. The finding that matters is the ordering instability on Amazon-VG. Those figures were seed-42 single runs, which cannot separate a protocol effect from run-to-run noise, so we repeated the contrast across the same ten seeds as the main results on MovieLens-1M. The effect is far larger than the noise: $v(\mathcal{G})$ rises from 0.05225 to 0.06880, a paired increase of $+31.7\%$ with a percentile-bootstrap interval of [$+30.2\%$, $+33.1\%$] and all ten seeds positive. The source ordering is nonetheless preserved on every seed ($\tau = 1.00$, 10/10, *seq* leading throughout). Per source, the increase is concentrated exactly where the mechanism predicts: *seq* gains $+0.01393$ [$+0.01369$, $+0.01418$] and *cf* $+0.00199$ [$+0.00155$, $+0.00247$], both with all ten seeds agreeing in sign, while *ct* and *rec* move by less than 10^{-4} with five seeds either way and no stable sign. A refreshed history helps the sequential source and essentially nothing else. The magnitude of the protocol effect is therefore established well

beyond seed noise, and so is the stability of the ordering under it.

Timestamp and metadata comparability All reported corpora carry genuine interaction timestamps: Gowalla spans 596 days with 99.2% distinct values and Amazon-VG spans 8360 days with 97.4%. This is necessary for *pop*, *rec*, and *seq* to have the declared temporal semantics. Two residual limitations remain. First, the content signal for *ct* and *rec* is derived rather than native on two corpora (geographic cells for Gowalla, category and store strings for Amazon-VG), so those players measure a defensible notion of item similarity, but not the curated metadata MovieLens provides. Second, exchangeability in the duplicate-source diagnostic validates symmetry independently of whether the cloned source has ideal domain semantics; it does not validate the absolute attribution magnitudes.

A candidate pool the players did not build The pool is coalition-independent, which is the property the game needs, but it is not source-independent: the five players construct it, so the pool is enriched for the items they rank highly. We therefore rebuilt the whole game on pools that consult no source score. Three rules are implemented. The *popularity* rule takes the $N_{\max}$ globally most frequent training items, the same list for every user; it is neutral with respect to four players but correlated with *pop*, which we state rather than gloss. The *random* rule draws $N_{\max}$ eligible items per user by a keyed hash, so it is strictly independent of every source and reproducible across platforms. Because recall then collapses to roughly $N_{\max}/|I|$, almost no user retains a reachable target and the game flattens for the dilution reason of Eq. (10), which would confound a null result with a null game; the *oracle* rule therefore restores the test item while leaving the distractors source-blind, and it is the cleanest of the three. Only orderings and signs are comparable across pools, since each rule yields a different $|C_u|$ and hence a different baseline and recall ceiling.

The attribution is largely, but not entirely, pool-independent. Kendall τ against the union pool is 0.80 for every rule on MovieLens-1M, 0.80, 0.80 and 0.60 on Amazon-VG, and 1.00, 0.80, 0.80 on Gowalla. The leading source is preserved under all three neutral rules on MovieLens and under the popularity rule everywhere; it changes on Amazon-VG under the oracle rule (*cf* $\rightarrow$ *seq*) and on Gowalla under both random rules (*cf* $\rightarrow$ *ct*, the near-tie already flagged in Section 4.3). Sign agreement holds in eight of the nine comparisons. The material disagreement itself is the most robust feature: *cf* on MovieLens and *pop* on Gowalla remain flipped under pools that no source helped build, so the phenomenon is not an artefact of letting the players choose the candidates.

One measurement here is an independent confirmation of Eq. (10) rather than a sensitivity result. Under the oracle rule every test item is inserted whenever it is eligible, so recall should equal ρ exactly. It does: 1.000 on MovieLens-1M, 0.972 on Amazon-VG and 0.501 on Gowalla, matching

the retrievability ceilings of Table 4 to three decimals. A pool that inserts the target by construction still cannot retrieve a target the training mask has removed.

Reproducibility of the recency clusters The *rec* cluster assignment is an argmax over SVD components, which is not invariant to a component sign flip, so the implementation now canonicalises the signs explicitly and we test that *rec* is bit-identical across solver seeds (Section 3.4). Investigating this produced a negative result worth recording: the convention we imposed is the one scikit-learn already applies internally, so the signs were never the operative hazard and no reported number changes. The residual risk is different in kind and cannot be fixed by a sign convention. Where two singular values coincide the corresponding subspace has no preferred basis, different solvers return different rotations of it, and the partition genuinely moves. That is a property of the spectrum, not of the seed, and it is the caveat that stands.

Metric sensitivity on MovieLens-1M NDCG@10 is the only metric in the characteristic function, so the attributions could in principle be an artefact of that choice. We re-scored the same game with Recall@10 and with MRR@10 in place of NDCG@10 on MovieLens-1M (seed 42), recomputing all 32 coalitions in each case. The induced source ordering is identical under all three (Kendall $\tau = 1.00$ against NDCG in both cases, *seq* first and *cf* second throughout). Magnitudes are not comparable, because the Recall and MRR games are not baseline-centred, so we report the ordering only. This is one corpus and one seed.

The full suite is uneven across corpora Robustness sweeps (Figure 4), segment analysis, and fusion are strongest on MovieLens-1M, the only corpus that clears the candidate-recall gate. Gowalla required a candidate pool of 11 623 items and still reaches only 0.450 recall, against an attainable ceiling of 0.501 imposed by its repeat-visit structure rather than by the pool (Section 3.5). Its absolute attribution magnitudes are diluted by that same factor and are the least comparable of the three, so we report Gowalla for the relative LOO-versus-Shapley contrast only, which Eq. (10) shows is exactly invariant to the dilution.

4.7. Which estimand, and does attribution predict retirement cost?

Two experiments test whether the paper's framing survives contact with the decision it motivates.

Three estimands Our main game refits the fusion head per coalition on a candidate set built from *all* sources. Two alternatives isolate what that choice contributes: v_{fixed} holds the deployed grand-coalition head $w^{(G)}$ fixed and merely masks it to S ; v_{e2e} lets each coalition retrieve its own $C_u(S)$ and refits its head. For $S \neq \emptyset$ and $u \in \mathcal{U}_{\text{eval}}$, define

$$v_u^{\text{fixed}}(S) = \text{NDCG@K} \left(\text{rank}(Z_u^{(S)} w_S^{(G)}); \mathbf{y}_u^{\text{te}} - b_u \right), \quad (16)$$

Table 11

Three estimands compared at seed 42, an ordering sensitivity rather than an inferential result. The refitted-head rows are the final source-symmetric main results. The fixed-head and end-to-end rows are legacy single-seed diagnostics from before symmetric-rule regeneration and are labelled accordingly. τ is Kendall agreement with the main game's ordering.

Corpus	Game	cf	ct	pop	rec	seq	τ
MovieLens-1M	refitted head (main)	+0.01545	−0.00009	+0.00637	+0.00026	+0.02937	n/a
	fixed head (legacy)	+0.01867	−0.00104	+0.00528	−0.00079	+0.02924	1.00
	end-to-end (legacy)	+0.01478	−0.00010	+0.00783	−0.00022	+0.02908	0.80
Amazon-VG	refitted head (main)	+0.01926	+0.00187	+0.00157	−0.00047	+0.01909	n/a
	fixed head (legacy)	+0.02110	+0.00101	+0.00166	−0.00045	+0.01799	0.80
	end-to-end (legacy)	+0.01962	+0.00133	+0.00291	+0.00026	+0.01718	0.80

$$r_u^{e2e}(S) = \text{rank}(Z_u^{(S)}(S)w^{(S)}), \quad (17)$$

$$U_u^{e2e}(S) = \begin{cases} \text{NDCG@}K(r_u^{e2e}(S); \mathbf{y}_u^{\text{te}}(S)), & |C_u(S)| > 0, \\ 0, & |C_u(S)| = 0. \end{cases} \quad (18)$$

$$v_u^{e2e}(S) = U_u^{e2e}(S) - b_u(S). \quad (19)$$

Here $b_u(S)$ is Eq. (7) evaluated on $C_u(S)$ and is zero when that set is empty. Aggregate values average these per-user quantities over the fixed population $\mathcal{U}_{\text{eval}}$. A removed source therefore loses its retrieval role too, and coalition values are no longer comparable across a fixed item set, which is exactly the property the main game is designed to preserve. Both alternatives take $v_{\text{fixed}}(\emptyset) = v_{e2e}(\emptyset) = 0$ by convention, matching the main game: the formulas above are not evaluated at $S = \emptyset$, where the fixed head would rank on an all-zero score vector and $C_u(\emptyset)$ would be empty.

Because $b_u(S)$ varies with the coalition, the retirement losses we report are differences of the *raw* U_{e2e} , per Equation (2). Differencing the centred v_{e2e} instead would silently add $b(\mathcal{G} \setminus \{g\}) - b(\mathcal{G})$ to every reported cost, so we computed both. The contaminating term is small here, at most 1.7×10^{-4} in absolute value on the pilot corpus, and it changes no ordering: raw and centred losses agree at Kendall $\tau = 1.00$ and select the same lowest-removal-loss source, so the reported retirement conclusions are unaffected. Both columns and the baseline shift are recorded so this distinction is auditable. Table 11 gives the three attribution vectors side by side rather than only their rank agreements. Its refitted-head rows are the final source-symmetric seed-42 main results used elsewhere in the paper; the fixed-head and end-to-end rows are explicitly labelled legacy single-seed diagnostics because those alternatives were not regenerated after the truncation-key audit. The table is therefore an ordering sensitivity, not a same-artifact equality check. All three games rank *seq* first on MovieLens-1M and *cf* first on Amazon-VG, so the ranking-stage restriction is consequential in the values but does not change the top-ranked source on either corpus. Gowalla is absent: the end-to-end game rebuilds candidates for each of the 2^n coalitions, which on an 82 134-item catalogue is 32 retrieval passes rather than one, and we did not run it.

Retirement simulation, and a negative result The stronger test is whether attribution predicts what removing a source actually costs. We removed each source *entirely*, from retrieval and fusion alike. We then rebuilt candidates from the survivors, refitted the head, and measured the observed end-to-end NDCG@10 loss under the reported protocol. This is one measured realisation, not a population-level “true” cost.

Which LOO is being tested The LOO column of Table 12 is LOO_{rank} : it comes from the same fixed-candidate game as the Shapley column, so the comparison is like for like. It is *not* LOO_{e2e} , which would equal the observed loss by construction (Equation 2) and could not be evaluated as a predictor of it. The result below is therefore not the tautology it might appear to be. It says something more useful: a quantity computable *without* re-running retrieval predicts the expensive end-to-end outcome well, and predicts it far better than the Shapley value of the same game.

LOO_{rank} predicts retirement cost better than Shapley (Table 12). On seed 42 its ranking matches the observed loss exactly ($\tau = 1.00$ against 0.20; exact two-sided permutation $p = 0.017$ and $p = 0.82$ at $n = 5$ sources), and it identifies *cf* as the source with the smallest observed removal loss, whereas Shapley selects *ct*. Removing *cf* improves the end-to-end metric by 0.00468. We write *observed* rather than *true* loss throughout: this is a measured realisation under the reported protocol, not a population quantity.

The pattern is stable across training seeds on the three fixed corpora Over ten seeds (Table 13) the picture is both broader and slightly more measured. LOO_{rank} 's mean rank agreement is 0.96, 0.96 and 1.00; Shapley's is 0.20, 0.74 and 0.80. LOO_{rank} is therefore *not* perfect on every seed, but has higher agreement on every corpus. The practical gap is starker than the correlations suggest: LOO_{rank} identifies the source with the smallest observed removal loss on 100%, 90% and 100% of seeds, against 0%, 50% and 0% for Shapley. On two of three corpora Shapley never once picks the right source to switch off. Ten seeds bound these proportions loosely, and we give Wilson intervals in Table 13 rather than let the point estimates stand alone: 100% is [72, 100] and 0% is [0, 28], so the separation on MovieLens and Gowalla

Table 12

MovieLens-1M source-retirement loss at seed 42. Negative loss denotes an NDCG@10 improvement after removal.

Source	Observed loss	Recall after	Shapley	LOO _{rank}
<i>cf</i>	−0.00468	0.691	+0.01545	−0.00480
<i>rec</i>	−0.00128	0.782	+0.00026	+0.00033
<i>ct</i>	−0.00050	0.785	−0.00009	+0.00115
<i>pop</i>	+0.00253	0.720	+0.00637	+0.00120
<i>seq</i>	+0.01702	0.659	+0.02937	+0.01672
Rank agreement with observed loss			$\tau = +0.20$	$\tau = +1.00$

Table 13

Retirement agreement over ten training seeds. Intervals quantify training-run variation on each fixed corpus. Paired $\Delta\tau$ is the effect measure; p_{Holm} adjusts the three paired Wilcoxon comparisons as one primary family.

Corpus	τ LOO _{rank}	τ Shapley	Paired $\Delta\tau$	p_{Holm}
MovieLens-1M	0.96 [0.90, 1.00]	0.20 [0.20, 0.20]	+0.76 [+0.70, +0.80]	0.0060
Amazon-VG	0.96 [0.90, 1.00]	0.74 [0.68, 0.80]	+0.22 [+0.16, +0.28]	0.0060
Gowalla	1.00 [1.00, 1.00]	0.80 [0.80, 0.80]	+0.20 [+0.20, +0.20]	0.0060

Corpus	LOO _{rank} correct [Wilson]	Shapley correct [Wilson]
MovieLens-1M	100% [72, 100]	0% [0, 28]
Amazon-VG	90% [60, 98]	50% [24, 76]
Gowalla	100% [72, 100]	0% [0, 28]

survives the interval while the Amazon contrast, [60, 98] against [24, 76], overlaps.

The comparison paired by seed For a conditional comparison of run-to-run stability, $\tau_{\text{LOO}_{\text{rank}}}$ and τ_{Shapley} are computed on the *same* seed, from the same fitted game, against the same observed retirement loss, so the seed is a block and the paired difference $\Delta\tau_s = \tau_{\text{LOO}_{\text{rank},s}} - \tau_{\text{Shapley},s}$ is both the correct statistic and a tighter one. It favours LOO_{rank} on every corpus: $\Delta\tau = +0.76 [+0.70, +0.80]$ on MovieLens-1M, $+0.22 [+0.16, +0.28]$ on Amazon-VG, and $+0.20$ with zero variance on Gowalla, where both correlations are seed-constant. No interval contains zero. Wilcoxon signed-rank over the ten paired seed differences gives raw $p = 0.0020$, 0.0039 , and 0.0020 . Holm correction over this declared three-comparison family gives $p_{\text{Holm}} = 0.0060$ for every corpus, conditionally on treating seeds as exchangeable runs. On Gowalla all ten differences equal $+0.20$, giving the minimum exact two-sided value $2/2^{10}$. The percentile intervals use 10000 resamples of ten seed-level differences; they describe training randomness on the fixed datasets and are not calibrated population intervals.

The result is also consistent with the theory rather than a contradiction of it. Removing *cf* improves end-to-end NDCG@10 by 0.00468 while dropping candidate recall from 0.748 to 0.691: the collaborative signal is substantially substitutive with *seq* and *pop* under the declared game (the pairwise interaction indices are -0.054 for *cf*–*seq* and -0.025 for *cf*–*pop*, the two most negative pairs), so in the grand coalition its scores add noise the others already cover. LOO_{rank} measures the removal of *cf* from

the fixed-candidate ranking game, and LOO_{e2e} measures it from retrieval and ranking together; the latter is the quantity aligned with observed offline single-source removal under this protocol, and the former tracks it closely here without re-running retrieval. Shapley averages over coalitions in which *cf* has no substitute, and therefore assigns the capability positive value *under the declared game*. We say that rather than “correctly”, because no external ground truth for the allocation exists: the two statements describe different games and need not agree. What is clear is that neither is the question a retirement decision asks.

The two methods therefore answer different questions. LOO_{e2e} is definitionally aligned with “what happens in this offline protocol if I remove this one source and keep the rest?”. Shapley answers the descriptive question “how should the system’s measured ranking quality be divided among its sources under the declared game?”. We claim only the latter, and we are deliberate about what it does not licence: a descriptive allocation of baseline-centred ranking utility is not a prescription for future engineering investment, which additionally depends on improvement headroom, implementation cost, latency, maintenance burden and deployment risk, none of which enter the characteristic function. Table 14 records that gap rather than papering over it.

5. Discussion

Which quantity answers which question The paper’s central practical result is that source allocation and source removal are different estimands, and that substituting one for the other can produce materially different decisions in

Table 14
Decision-to-estimand mapping.

Decision	Quantity aligned with question	Evidence in this study
Remove one deployed source	LOO_{e2e} by definition; LOO_{rank} a measured proxy here	$\tau = 0.96\text{--}1.00$ over 10 seeds (Sec. 4.7)
Divide baseline-centred ranking utility	Shapley under the declared game and axioms	efficiency and symmetry checks (Sec. 4.1)
Rank sources under a different coalition weighting	a declared semivalue (Banzhaf, binomial)	ordering agrees on MovieLens (Sec. 4.2); these do <i>not</i> sum to $v(\mathcal{G})$, so they rank rather than divide
Locate pairwise interaction under the game	pairwise interaction index	Table 7
Decide where to invest next	<i>not established</i>	would need cost and headroom, which v does not contain

the evaluated systems. Table 14 states the mapping we can defend from the evidence here.

What a positive Shapley value does and does not license When SignalShap assigns collaborative filtering $+0.01573$ on MovieLens while LOO_{rank} assigns -0.00457 in the ten-seed means of Table 8, the correct reading is not that ablation is mistaken. It is that under the declared ranking-stage game, *cf* contributes in the coalitions where its substitutes are absent, and that removing it from the deployed system nevertheless improves the metric because those substitutes are present. Both statements are true of the same system. An allocation is a statement about a game; an intervention is a statement about a deployment.

The limits of the allocation reading We motivate allocation by analogy with dividing engineering effort, and we should be honest that this analogy is untested. A source with high current allocated value is not necessarily the source where additional work pays best: that depends on marginal improvement potential, engineering cost, latency, and maintenance burden, none of which enter v . We therefore present the allocation as a description of ranking utility above the declared random baseline in the current fitted system, not as a prescription for the roadmap, and we regard closing that gap, an attribution whose payoff includes cost, as the most useful next step for this line of work.

On the strength of the retirement result Under the evaluated five-source protocol, ranking-stage LOO has higher agreement with observed retirement cost on each fixed corpus over ten stochastic fits. With five sources the exact two-sided permutation p for $\tau = 1.00$ is 0.017, so a single corpus and a single seed would be suggestive rather than conclusive. Three corpora and ten stochastic fits make the pattern more stable to training randomness, but do not provide independent data-level replications. The decision-level figures are more striking than the correlations: on MovieLens and Gowalla, Shapley never once selects the source with the smallest observed removal loss. It remains a result about five-source hybrids under one deployment protocol, and we would not extrapolate it to systems with many more

components, where the grand coalition is a less special point in the lattice.

5.1. Threats to validity

Construct validity: what the game measures The candidate set is the union of *all* sources' top- N lists and is held fixed across coalitions. This is deliberate: it makes $v(\mathcal{S})$ comparable, since a coalition-specific pool would hand the grand coalition a retrieval advantage. Its consequence is that every coalition, including the empty one, ranks items that absent sources helped retrieve. SignalShap therefore estimates *ranking-stage* credit, not end-to-end subsystem value. Retiring a source in production would also remove its retrieved candidates, an effect this game excludes by construction. Claims about subsystem investment should be read with that scope in mind, and separating v_{rank} from a coalition-retrieval v_{e2e} is the most valuable extension we can identify.

Construct validity: what the validation establishes The duplicate experiment verifies symmetry and efficiency on real fitted games. It does not establish absolute correctness of the magnitudes, and no experiment here does, because no independent ground-truth attribution exists for these systems. Analytic games with a fully known Shapley vector complement this and are reported in Section 4.2: all six reproduce exactly, including the null-player and harmful cases that the symmetry diagnostic cannot reach.

Statistical conclusion validity The inferential target is conditional algorithmic stability under each fixed corpus, split, and user sample. The study does not define a sampling population of corpora or users from which the three systems or ten seeds are independent draws, so its intervals and tests do not support population prevalence claims. All users share the same fitted source models, so treating users as independent understates dependence. The same users also appear under every seed, so users and seeds form a crossed rather than a nested design. The exploratory two-level bootstrap in Appendix A resampled users independently within each seed and therefore does not preserve that pairing. Its interval is

reported as a sensitivity diagnostic, not as a calibrated confidence interval or a basis for a positive equivalence claim. A confirmatory analysis requires a multiway bootstrap, or common user resamples across seeds, together with both one-sided TOST statistics and a 90% equivalence interval.

The t_0 intervals elsewhere describe variation across ten stochastic fits on one fixed dataset and split. They do not quantify sampling uncertainty over users or corpora. Kendall τ is bounded and discrete at five sources, so its bootstrap intervals are coarse and can be degenerate when every seed returns the same ordering. The paired retirement comparison remains useful as a stability result, but its Wilcoxon p -values are conditional on treating the ten seeds as exchangeable training runs. Likewise, user-level tests against uniform, popularity, LightGCN and SASRec are descriptive only: a large effect does not restore independence, and the neural tuning budgets are unmatched.

On Amazon-VG, a user-level test gives corrected $p = 0.019$ for fusion versus the global head, but $d_z = 0.014$ and 6951 of 7120 users are exact ties. We report this only to illustrate why conditional user-level significance is not the evidential target.

Internal validity The coalition heads and the coalition values are fitted and evaluated on *disjoint* folds: the ridge head for each S is fitted against validation targets, and $v(S)$ is the NDCG of the resulting ranking against held-out test items. The main attribution result is therefore not in-sample, and a continuous-integration check asserts the separation so a refactor cannot reintroduce it. The attribution-derived fusion weights of Appendix A are a different matter: there the same validation fold supplies both the coalition heads and the attributions used to build the weights, which is an in-sample optimisation bias even though no test label is touched. Proper cross-fitting would be required to make that a positive claim. Since the observed difference from the simpler baseline is negligible, we report the experiment as a negative result rather than as evidence of equivalence or superiority. Baseline tuning budgets for LightGCN and SASRec were modest and are not matched to the effort spent on our own pipeline; the absolute figures for those models should be read as reference points under our protocol rather than as their attainable best.

External validity One subsample per corpus, drawn to fit a memory budget, means cross-corpus results may be sample-specific. The three corpora vary in density, catalogue size, metadata provenance, and sampling simultaneously, so this is not a controlled density comparison; the density ordering is clean (0.075%, 0.469%, 2.697%, a 36 $\times$ spread with no adjacent pair closer than 5.7 $\times$), but the corpora are not otherwise matched, and we do not attribute the differences between them to density alone. The study also does not evaluate user cold start, item cold start, graded relevance, or online behavior, and its players are deliberately simple rather than current graph or Transformer recommenders.

The results establish an estimand distinction in these five-source hybrids, not performance or attribution validity for modern production architectures generally.

6. Conclusion

We formulated hybrid-recommender source attribution as a five-player cooperative game and computed Shapley values by exact enumeration over all 32 coalitions, avoiding the estimator variance that often motivates sampling in model-agnostic feature attribution. We showed that a natural statement of the redundancy property is false without monotonicity, gave the counterexample, and audited the hypothesis rather than assuming it.

The strongest implementation evidence is the set of invariant checks. Injecting a duplicate into a real system creates a constraint that holds independently of the data, namely that exchangeable players must receive equal credit, and SignalShap satisfies it to numerical precision, alongside efficiency to 6.9×10^{-18} across all three corpora. These are unit and invariant tests of the Shapley aggregation, not validation that the declared game gives a uniquely correct real-world allocation. Under the same condition, ablation assigns zero credit to both substitutes.

We are explicit about what remains open, and about one question we answered against ourselves. These checks do not identify the absolute correct credit, and the game measures ranking-stage rather than end-to-end contribution. We also ran the experiment that bears most directly on use: in the single-source retirement decision studied here, ranking-stage leave-one-out had higher agreement with the ordering of observed end-to-end removal losses than ranking-stage Shapley on each fixed corpus over ten stochastic fits (Section 4.7). This is conditional case-study evidence, not population-level inference. It does not make leave-one-out a better general allocation rule; it shows that the estimand aligned with a removal decision suits that decision. We therefore claim credit allocation, not retirement guidance. Under the frozen primary configuration, only MovieLens-1M clears the candidate-recall gate; Amazon-VG also clears it in the candidate-cap sensitivity without changing the qualitative attribution result. Gowalla remains below the gate and is interpreted only as a repeat-masked relative stress test. Under the training-history mask, its candidate recall is bounded above by $\rho = 0.501$, so the 0.60 gate is unreachable at any candidate-pool size. Resolving this limitation requires a repeat-aware evaluation protocol rather than simply increasing $N_{\max}$.

Future work Four directions follow directly from what we could not settle. First, a cost-aware estimand: allocation describes the fitted system's utility above the declared baseline, but deciding where to invest needs marginal improvement potential and engineering cost inside the payoff, and neither is in v . Second, scaling beyond a handful of players, where exhaustive enumeration fails and the exactness that motivates this design must be traded against sampling error. A

comparison of exact and sampled values across n would establish where the boundary lies. Third, corpora and primary candidate configurations for which fixed-candidate retrieval clears the recall gate. Fourth, richer players: graph-neural and Transformer components, privacy-bearing signals, and cascade rather than parallel hybrids, all of which change the retrieval structure the game assumes.

A. Attribution-Derived Fusion: A Negative Result

This appendix reports an experiment that did not work. We include it because the natural next step after allocating credit is to *use* that allocation, and a reader who asks whether attribution-derived weights help deserves the measured answer rather than silence. They do not.

Two questions separate here: whether attributions *differ* across user segments, and whether exploiting that difference *pays*.

Segments and the weight map Segments are defined explicitly as a 2×2 split on training behaviour only, computed before any test data is touched: activity (above or below the median interaction count) crossed with recency skew, the fraction of a user's interactions falling in the newer half of that user's own timespan, again split at the median. This yields four segments; users with no training interactions fall in the first.

Segment attributions are mapped to fusion weights by a softmax with shrinkage toward the global profile. For segment s and source g ,

$$q_{s,g} = \frac{\exp(\varphi_{s,g}/\tau)}{\sum_{h \in \mathcal{G}} \exp(\varphi_{s,h}/\tau)}, \quad (20)$$

$$q_{\text{global},g} = \frac{\exp(\varphi_{\text{global},g}/\tau)}{\sum_{h \in \mathcal{G}} \exp(\varphi_{\text{global},h}/\tau)}, \quad (21)$$

$$a_{s,g} = (1 - \gamma_{\text{shrink}}) q_{s,g} + \gamma_{\text{shrink}} q_{\text{global},g}. \quad (22)$$

The softmax is not cosmetic. Our games are non-monotone, so $\varphi_{s,g}$ can be negative, and using raw values as weights would *invert* a source's scores rather than down-weight it, a qualitatively different operation from the one the attribution licenses. Shrinking a noisy per-segment estimate toward a pooled one is the classical variance-reduction trade [46]. The temperature τ and shrinkage γ_{shrink} are selected on the validation fold and frozen before any test evaluation.

Segment heterogeneity is descriptive, not established On MovieLens-1M, 7 of 30 segment-pair contrasts are significant at $\alpha = 0.05$ uncorrected, against 1.5 expected by chance, and Amazon-VG gives the same 7 of 30. Gowalla gives 4 of 30, but that count is still under the legacy candidate rule (Table 10), because regenerating this corpus under the symmetric rule exhausted memory at 8865×82134 . All three are above the chance rate, and after Holm correction within each corpus none of the three corpora retains a significant contrast. We therefore make no confirmatory segment-heterogeneity claim. The profiles in Figure 6 remain useful

for showing the scale of the variation that the downstream fusion experiment attempted to exploit.

No cross-corpus omnibus ranking A cross-corpus Friedman test would pool absolute NDCG@10 from two corpora that fail the declared recall gate under the frozen primary configuration and would use only three corpus blocks. We therefore do not report or interpret an omnibus method ranking; the appendix is descriptive per corpus.

Fusion weights never see test outcomes Because the characteristic function of Section 3.1 is evaluated on test interactions, deriving fusion weights from those same attributions would be circular. We avoid this: the fusion heads, the segment-specific weights, the choice of head family, and the shrinkage strength are all fitted on the *validation* fold only, and are frozen before a single test evaluation. The Shapley values reported in Section 4.3 are diagnostic output; they are not inputs to the weights.

Attribution-derived fusion gives no resolvable gain over the matched head On the MovieLens illustration, weights derived from segment-level attributions are numerically above uniform fusion ($+0.0118$, $d_z = 0.097$), a popularity reference ($+0.0452$, $d_z = 0.232$), LightGCN [47] ($+0.0295$, $d_z = 0.157$), and SASRec [48] ($+0.0464$, $d_z = 0.243$), at NDCG@10 = 0.06348 (Figure 7). These are descriptive differences, not superiority tests: users share fitted models, the neural references received smaller tuning budgets, and every effect size is below 0.25.

Against a globally tuned head of the same family (NDCG@10 = 0.06357) the attribution-derived head is 0.00009 *lower* ($d_z = -0.007$, Holm-corrected $p = 0.23$), which is practically negligible and, in this run, negative. Under the legacy candidate rule the same comparison came out at $+0.00003$: the difference is not stable in sign across the candidate construction, and it is not stable under the plausible per-user normalisation in Eq. (8) either, while its magnitude stays within noise throughout. A quantity whose sign depends on the truncation key is not evidence of an advantage in either direction. **We therefore make no claim of any advantage over a well-tuned global head**, and note that the comparisons against LightGCN and SASRec are conditional on tuning budgets that were not matched to our own pipeline.

An exploratory bootstrap that resamples seeds and then users within each seed gives a mean difference of $+0.00005$ with interval $[-0.00022, +0.00032]$. Because the same users occur under every seed, that nested procedure does not preserve the crossed pairing and is not used for a positive equivalence claim. A confirmatory equivalence analysis would require a multiway or pairing-preserving bootstrap, a prespecified equivalence margin, a 90% interval, and both one-sided statistics. The defensible conclusion is narrower: no advantage over the matched global head is resolved in these data.

The contribution stands on the implementation checks of Section 4.1 and the attribution results of Section 4.3. This

Exact-enumeration ranking-stage source attribution

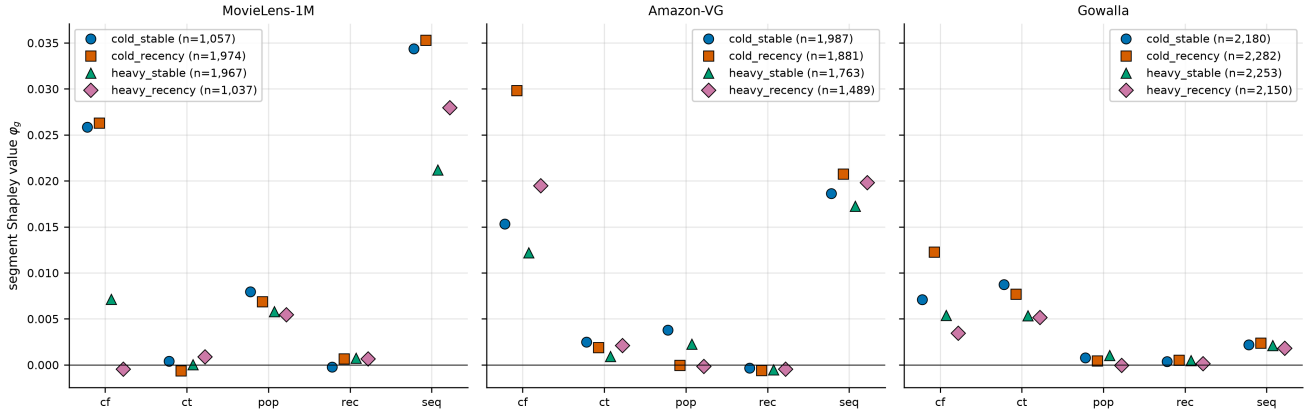


Figure 6: Segment-level Shapley values; legends report segment sample sizes.

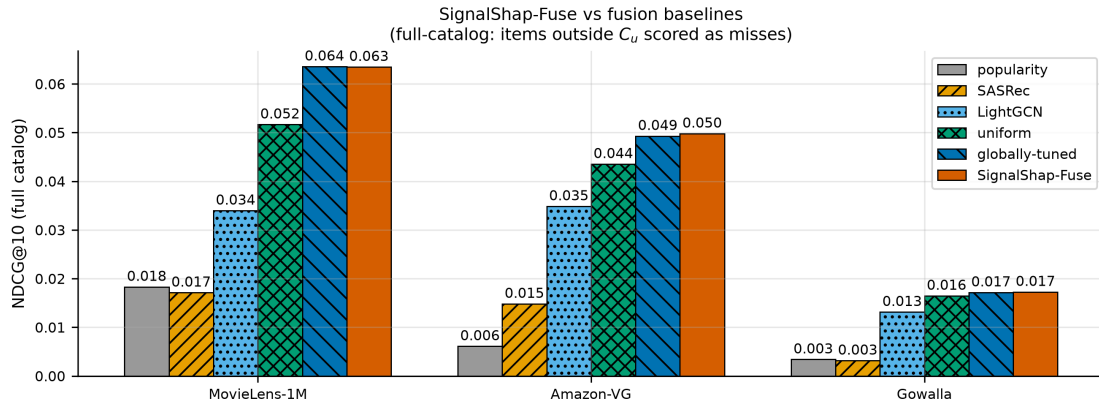


Figure 7: Fusion comparison under full-catalogue evaluation. Under the frozen primary configuration, Amazon-VG and Gowalla remain below the declared candidate-recall gate; Amazon-VG clears it in the candidate-cap sensitivity.

appendix is a negative result about one downstream use of the attributions, and it is not evidence either for or against their correctness.

CRedit authorship contribution statement

Mouad Louhichi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Redwane Nesmaoui:** Software, Validation, Data curation, Writing – review and editing. **Mohamed Lazaar:** Supervision, Methodology, Resources, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This work uses three public corpora and redistributes none of them. MovieLens-1M is used under the GroupLens research-use license, Gowalla check-ins come from the SNAP collection, and the Video Games category is taken from the Amazon Reviews 2023 corpus. The corpora themselves are not redistributed because the GroupLens license forbids it.

Code availability

The implementation, configuration files, and scripts required to reproduce the reported experiments, tables, and figures are publicly available at <https://github.com/mouadlouhichi/signalshap.git>.

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